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Length Generalisation in Transformer-Based Sequence Models: Failure Taxonomy and Mitigation Strategies

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Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

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I would also like to acknowledge the University of Bristol for providing the resources and academic environment that made this work possible.

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Supporting Technologies

- Python was used to implement dataset generation, model training, evaluation, diagnostic analysis, and result visualisation.
- PyTorch was used to implement the Transformer models, positional encoding variants, optimisation procedure, checkpointing, and GPU execution.
- NumPy and pandas were used for numerical processing, aggregation of experimental results, and construction of reproducible summary tables.
- Matplotlib was used to produce the dissertation figures. A common visual style, colour mapping, typography, and output format were applied across the final plots.
- PyYAML configuration files were used to define experiment settings separately from the training and evaluation implementation.
- The aeon dataset tools were used to obtain and prepare the publicly available SelfRegulationSCP2 time-series dataset.
- Amazon Web Services EC2 G5-family GPU resources were used for the multi-seed training and evaluation runs that were impractical to execute locally.
- Git and GitHub were used for version control and for synchronising the experimental implementation between local and AWS environments.
- LaTeX was used to prepare the dissertation and to include publication-quality PDF figures.

Notation and Acronyms

Acronyms

ALiBi Attention with Linear Biases.

AWS Amazon Web Services.

CSV Comma-separated values.

CUDA Compute Unified Device Architecture.

EEG Electroencephalography.

EM Exact match.

FFN Feed-forward network.

GPU Graphics processing unit.

JSONL JSON Lines.

LSTM Long short-term memory.

NoPE No explicit positional encoding.

OOD Out-of-distribution.

RNN Recurrent neural network.

RoPE Rotary Position Embedding.

SD Standard deviation.

UEA University of East Anglia.

YAML YAML configuration format.

Mathematical Notation

Q, K, V Query, key, and value matrices used in self-attention.

d_k Dimensionality of the key and query vectors within one attention head.

$B_{ij}^{(h)}$ ALiBi attention-score bias between query position i and key position j in attention head h.

m_h Fixed distance-bias slope used by ALiBi attention head h.

L A generic sequence or evaluation length.

L_{train} Sequence length used to train a fixed-length configuration.

L_{test} Sequence length used during evaluation.

L_{collapse} First evaluated unseen length at which the three-seed mean exact-match accuracy falls below the 10% collapse threshold.

n Index of an evaluated example.

j Index of a target-output position.

N Number of examples in an evaluation set.

$y_{n,j}$ Target token at position j in example n.

$\hat{y}_{n,j}$ Token predicted at position j in example n.

$m_{n,j}$ Binary evaluation mask: one for an evaluated target position and zero for padding.

$\mathbb{I}[\cdot]$ Indicator function, equal to one when its condition is true and zero otherwise.

$A_{\mathrm{tok}}(L)$ Token accuracy, expressed as a percentage, at evaluation length L.

$A_{\mathrm{EM}}(L)$ Exact-match accuracy, expressed as a percentage, at evaluation length L.

$\bar{A}_{\mathrm{EM}}(L)$ Mean exact-match accuracy across the three baseline seeds at evaluation length L.

$E_{\mathrm{tok}}(L)$ Token-error percentage at length L, $100 - A_{\mathrm{tok}}(L)$.

$E_{\mathrm{EM}}(L)$ Exact-match error percentage at length L, $100 - A_{\mathrm{EM}}(L)$.

E_b Token-error percentage within relative-position bin b.

$\Delta A_{\mathrm{EM}}(L)$ Difference in exact-match accuracy between an intervention and its matched original baseline at length L, measured in percentage points.

d_i Normalised input-output dependency distance associated with Reverse output position i.

Ethics Statement

This project did not recruit human participants, conduct interviews or surveys, or collect new personal data. The principal experiments used programmatically generated synthetic sequences for Addition, Delayed Copy, and Reverse. These tasks contain no information relating to identifiable individuals.

The exploratory real-world experiment used the publicly available SelfRegulationSCP2 dataset from the UEA Multivariate Time Series Classification Archive [2]. Although this dataset contains preprocessed EEG recordings, the project used only the distributed research dataset and its existing class labels. No attempt was made to identify participants, link records to external information, or draw clinical conclusions about individuals. Results are reported only as aggregate classification accuracies across positional encoding conditions, sequence lengths, and random seeds. The experiment is presented as a limited methodological extension and not as a diagnostic or clinical system.

Research integrity was supported by retaining experiment configurations, random seeds, raw result files, and analysis scripts. Negative and inconclusive outcomes are reported explicitly: configurations that failed to learn the training task are identified as underfitting, and interventions that preserved training performance but did not improve unseen-length accuracy are described as showing no observed improvement under the tested mitigation. Attention statistics are treated as descriptive evidence rather than causal explanations.

Computational resources were also considered. Compact Transformer models were used, checkpoints and result files were retained to avoid unnecessary retraining, and GPU execution was reserved for the larger experiment matrix and checkpoint-based diagnostics. The source code, configuration files, and reproduction instructions are provided through the repository guide documented in Appendix C.

Abstract

Transformer models have become a dominant architecture for sequence modelling, partly because self-attention allows them to represent dependencies across an input without the recurrence used in earlier sequence models. However, strong performance on the sequence lengths observed during training does not necessarily imply that the same model has learned a rule that generalises to longer sequences. This limitation is important because many practical sequence-modelling problems require models to operate reliably outside the exact length distribution seen during training.

This dissertation investigates length generalisation in Transformer-based sequence models using a controlled train-short/test-long experimental framework. The project focuses on three synthetic sequence tasks: addition, delayed copy, and reverse. These tasks were selected because they test different forms of sequence reasoning, including symbolic computation, long-range memory, and position-dependent transformations. A compact Transformer model was evaluated across five positional encoding settings: learned positional embeddings, sinusoidal positional encodings, no explicit positional encoding, ALiBi, and RoPE. Experiments were repeated across three random seeds to assess robustness rather than relying on single-run results.

A baseline-eligibility procedure first identified configurations that reliably learned the training task. Eligible configurations were then analysed using exact-match accuracy, token-error percentages, position-wise error profiles, attention diagnostics, and task-specific tests of carry conditions and dependency distance. Mixed-length training, curriculum learning, and randomised padding were evaluated against matched original baselines to test specific hypotheses about exposure to varied lengths and dependence on absolute positions.

All five eligible task-encoding configurations that met the multi-seed baseline-eligibility criterion fell below the 10% exact-match threshold at the first unseen length. At this point, Addition showed approximately 86.9–88.0% token error, with errors increasing modestly towards later output positions. Delayed Copy and Reverse showed approximately 98.8–99.0% token error that was almost uniform across output positions. Addition errors differed by less than one percentage point between digits with and without an incoming carry for the learned model and were effectively identical for the sinusoidal model. Reverse errors were not concentrated at a particular dependency distance. Mixed-length variants underfit in the evaluated settings, while curriculum and randomised-padding variants that retained training-length performance produced no observed improvement at unseen lengths. In the limited SelfRegulationSCP2 extension, mean classification accuracy remained approximately 49–56% across the evaluated input lengths rather than showing the catastrophic exact-match collapse observed on the synthetic tasks.

The resulting failure taxonomy distinguishes training-length underfitting, seed instability, immediate extrapolation collapse, position-dependent error structure, and no observed improvement under tested mitigation. These findings show that length-generalisation evaluation should establish a valid training baseline before examining not only whether performance decreases, but how failure is distributed and which intervention hypothesis was actually tested.

Keywords: Length generalisation; Transformers; positional encoding; synthetic sequence tasks; extrapolation; failure taxonomy; baseline eligibility; position-wise diagnostics; mitigation strategies; time-series classification.

Chapter 1

Introduction

1.1 Context and Motivation

Sequence models are often trained using inputs of a restricted length because of data availability, memory limitations, or computational cost, but may later be required to process substantially longer sequences. Reliable performance therefore requires more than fitting the length distribution observed during training: the model must transfer the learned computation to previously unseen lengths.

Transformers are widely used for sequence modelling because self-attention allows direct interaction between positions without the recurrent computation used in earlier architectures [28]. However, self-attention does not guarantee that the learned solution is independent of sequence length. A Transformer may achieve high accuracy on the training distribution while relying on absolute positions, fixed dependency distances, or patterns specific to the lengths encountered during training. Strong in-distribution performance can therefore conceal substantial failure when the same task is evaluated at longer lengths.

Length generalisation is the ability of a model trained on shorter instances to perform the same underlying task at longer, unseen lengths. It is a form of out-of-distribution generalisation in which the task rule remains unchanged while sequence length varies. Previous research has shown that this remains difficult for Transformers, including on algorithmic tasks whose rules should, in principle, apply at arbitrary lengths [1, 31]. This makes such tasks useful for distinguishing genuine rule learning from solutions that work only within the training-length distribution.

Positional encoding is closely connected to this problem because Transformers must represent order at positions that may not have appeared during training. Learned embeddings, sinusoidal encodings, rotary position embedding (RoPE), Attention with Linear Biases (ALiBi), and models without explicit positional encoding (NoPE) represent position in different ways [28, 24, 19, 11]. Randomised positional strategies have also been proposed to increase exposure to positions outside the ordinary training range [21]. Nevertheless, a positional representation that supports longer inputs does not necessarily cause the model to learn a length-general task solution. Length generalisation is therefore related to, but distinct from, increasing a model's usable context window or improving the efficiency of long-context processing [6, 26].

A further challenge is how failure should be evaluated. Accuracy at the longest test length alone cannot establish whether a model learned the training task, where deterioration began, or how errors were distributed within its output. A configuration that never learns the training length is an example of underfitting, not a valid demonstration of extrapolation failure. Similarly, two models with zero sequence-level exact-match accuracy may have different token-error rates and positional error structures. Understanding length generalisation therefore requires a valid baseline and diagnostics that describe both when and how failure occurs.

1.2 Research Problem, Aim, and Questions

This dissertation investigates length generalisation using a controlled train-short/test-long framework. Its central aim is to distinguish observable forms of failure in compact Transformer models and to determine whether targeted training interventions alter those behaviours. The study first establishes which configurations reliably learn the training task and then examines their behaviour at progressively longer lengths.

The primary research question is:

When a Transformer trained on short sequences is evaluated on longer sequences, which distinguishable failure behaviours occur, where within the output do errors arise, and do targeted training interventions alter these behaviours?

This question is addressed through five supporting questions:

1. Which task-encoding configurations reliably learn the training-length task across multiple random seeds?
2. At which unseen length does performance first collapse for eligible configurations, and how does it change thereafter?
3. How are errors distributed across output positions and task-specific conditions such as incoming carries and dependency distances?
4. How do the tested positional encodings differ in training fit, seed stability, and extrapolation behaviour?
5. Do mixed-length training, curriculum learning, or randomised padding improve unseen-length performance relative to matched baselines while preserving training-length performance?

1.3 Scope and Contributions

The main benchmark uses three deterministic synthetic tasks: addition, delayed copy, and reverse. These tasks test symbolic computation, long-range retention, and position-dependent transformation, respectively. Five positional encoding settings are evaluated across three random seeds: learned, sinusoidal, NoPE, ALiBi, and RoPE. Associative recall was explored during development but excluded from the principal comparison because it was not learned reliably at the training length.

The study evaluates exact-match degradation, token-error percentages, multi-seed stability, position-wise errors, attention behaviour, incoming-carry conditions for addition, and dependency distance for reverse. Mixed-length training, curriculum learning, and randomised padding are compared with matched original baselines. A limited SelfRegulationSCP2 time-series classification experiment is also included to contrast the synthetic sequence-generation results with a real-world long-sequence setting. This extension is exploratory and is not presented as a state-of-the-art time-series benchmark [16].

The principal contributions are:

- a reproducible multi-task and multi-seed benchmark that separates training-length fit from length extrapolation;
- a baseline-eligibility procedure that prevents underfitting or seed-dependent success from being misclassified as extrapolation failure;
- position-wise and task-specific diagnostics that identify where errors occur rather than reporting only final accuracy;
- matched comparisons of three training interventions, each evaluated according to the failure mechanism it was intended to address; and
- an evidence-based taxonomy distinguishing training-length underfitting, seed instability, immediate extrapolation collapse, position-dependent error structure, and no observed improvement under the tested mitigation.

1.4 Dissertation Structure

Chapter 2 reviews the technical background and related literature, and Chapter 3 describes the experimental methodology and controls. Chapter 4 combines the synthetic benchmark results, failure diagnostics, mitigation experiments, taxonomy, and exploratory real-world extension. Chapter 5 interprets these findings in relation to the research questions and discusses their implications and limitations. Chapter 6 summarises the conclusions, contributions, stage reached relative to the project aim, and directions for future work. Supplementary results and methodological documentation are provided in Appendices A and B.

Chapter 2

Background and Related Work

This chapter introduces the technical and research context required to understand the experiments in this dissertation. It first summarises the Transformer components relevant to the implementation, then compares the positional encoding methods studied in the project. It subsequently defines length generalisation, relates it to controlled algorithmic tasks, and reviews the principles used to distinguish different forms of failure. The chapter concludes by examining relevant training interventions and identifying the research gap addressed by this work.

2.1 Transformer Sequence Modelling

Sequence-to-sequence models learn a mapping from an ordered input to an ordered output. Earlier neural approaches commonly used recurrent architectures, which process tokens sequentially and maintain a hidden state [25]. Attention mechanisms were later introduced to allow models to select information from different input positions rather than compressing the complete input into a single fixed representation [3].

The Transformer replaces recurrence with self-attention [28]. Given query, key, and value matrices Q, K, and V, scaled dot-product attention is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{2.1}$$

where d_k is the dimensionality of each key vector. The similarity between a query and the available keys determines how strongly their corresponding values contribute to the output. Scaling by $\sqrt{d_k}$ prevents large dot products from producing excessively concentrated softmax distributions.

Multi-head attention performs this operation in several representation subspaces. Different heads may therefore capture different relationships between positions. A Transformer block combines multi-head attention with a feedforward network, residual connections, layer normalisation, and dropout. Stacking these blocks allows information to be transformed using both token-level representations and interactions across the sequence.

The synthetic experiments in this dissertation use a compact, encoder-style Transformer in which every position can attend to every other position. The purpose is not to reproduce the scale of a modern language model, but to provide a controlled architecture that can be held constant while task, sequence length, positional encoding, and random seed are varied. Full architectural and optimisation settings are reported in Chapter 3.

2.2 Positional Information in Transformers

Self-attention compares token representations without inherently assigning meaning to their order. Positional information must therefore be introduced if the task depends on where a token occurs. This can be done by modifying the input representations or by changing the attention calculation itself.

The closest empirical studies reach different conclusions because they use different architectures, objectives, task representations, and training distributions. Table 2.1 compares the evidence most directly relevant to the present experiments.

Study and setting	Training and evaluation	Reported evidence	Boundary relevant here
Press et al. [19]; causal language modelling with ALiBi	Models trained at lengths 512 or 1024 and evaluated at 1024 or 2048	ALiBi supported lower-perplexity evaluation at contexts longer than those used during training.	The objective, causal attention regime, model scale, and perplexity metric differ from the present compact bidirectional models and deterministic exact-output tasks.
Kazemnejad et al. [11]; decoder-only Transformers on reasoning and mathematical tasks	Task-specific short training lengths followed by longer held-out instances	NoPE frequently outperformed explicit positional methods, while the effect of positional encoding and scratchpad format varied across tasks.	The causal architecture and task representations differ from those used here, so the result does not predict that NoPE must fit every controlled task.
Ruoss et al. [21]; 15 algorithmic tasks and 6,000 trained models	Short sequences were assigned ordered indices sampled from a wider positional range before evaluation on longer sequences.	Randomised positional encoding improved unseen-length accuracy by 12.0% on average.	The intervention directly randomised positional indices; the randomised-padded condition tested here changes token placement but does not implement that method.
Zhou et al. [31]; decoder-only Transformers on algorithmic tasks	Task-specific short training instances followed by longer held-out instances	Length generalisation correlated with the existence of a comparatively simple length-independent computational description, and scratchpad format could help or hinder it.	The present dissertation diagnoses empirical failure behaviour but does not identify or prove the internal algorithm learned by a model.

Table 2.1: Comparison of prior train-short/test-long evidence and the experimental differences that limit direct transfer of its conclusions to the present study.

Learned absolute positional embeddings are flexible within the range represented during training, but extrapolation is not built into their parameterisation. Fixed sinusoidal encodings instead use deterministic functions of position and can be evaluated beyond the training range [28]. Being mathematically defined at a new position, however, is different from the model having learned how to use that position correctly.

Relative positional approaches represent relationships between positions rather than assigning each location an independent learned vector [22]. RoPE rotates query and key representations as a function of position, allowing relative offsets to influence their dot products [24]. ALiBi applies head-specific linear biases to attention scores according to distance and was introduced explicitly in a train-short/test-long setting [19]. These designs provide mechanisms that can operate at longer lengths, but neither mechanism alone guarantees that the remaining network learns a length-general computation.

NoPE removes the explicit positional component, and prior work has reported useful length-generalisation behaviour for NoPE Transformers in particular decoder-only settings [11]. ALiBi was introduced in a causal language-modelling setting and demonstrated train-short/test-long behaviour under that architecture and objective [19]. Randomised positional encoding has likewise improved extrapolation by sampling position indices from a range wider than the ordinary training sequence length [21].

These findings are not directly interchangeable. They differ in attention regime, training objective, task representation, architecture, and the way positional information is introduced. The present experiments instead use a compact bidirectional encoder and deterministic sequence-to-sequence tasks. Consequently, the results test whether the reported positional mechanisms transfer to this controlled setting; they do not constitute direct replications or universal assessments of the original methods.

The comparison therefore separates two questions: whether a positional condition permits reliable learning at the training length, and whether the resulting solution transfers to unseen lengths. This distinction is necessary because training-length underfitting provides no evidence about how a successfully trained model would extrapolate.

2.3 Length Generalisation

Length generalisation concerns performance when

$$L_{\text{test}} > L_{\text{train}},$$

while the underlying task rule remains unchanged. The change is therefore in the size of the problem instance rather than in the rule being applied. A model demonstrates length generalisation only if it first learns the task within the training distribution and then applies that solution successfully at unseen lengths.

Strong interpolation performance can coexist with failure under compositional or length shifts [12, 27]. Formal-language and algorithmic studies similarly show that Transformer generalisation depends on task structure, representation, training distribution, and architectural assumptions [5, 8, 23, 31].

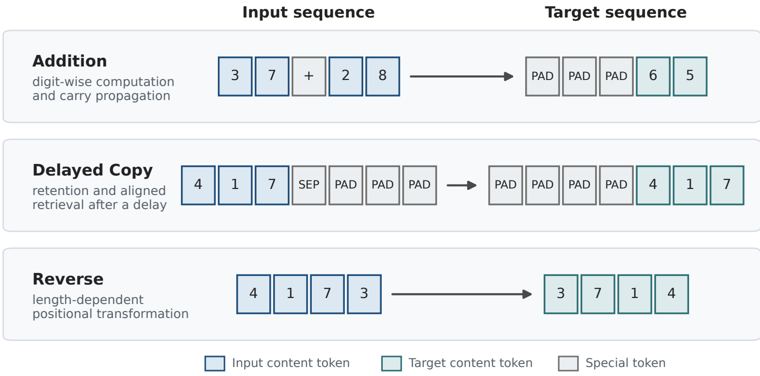

Figure 2.1: Representative input–target mappings for the three controlled tasks. Addition combines digit-wise computation with carry propagation; delayed copy requires retrieval after a separator and padded delay; and reverse applies a length-dependent positional transformation. `PAD` and `SEP` denote padding and separator tokens.

Controlled tasks keep the underlying rule fixed while sequence length changes. Addition tests digit computation and carry propagation, Delayed Copy tests retrieval after a delay, and Reverse tests position-dependent remapping. Together, they sample computation, retention, and positional transformation. Prior arithmetic and compositional studies show that representation, task formulation, and architectural choices can substantially alter extrapolation behaviour [13, 15, 17]. Exact task serialisations are given in Chapter 3.

Figure 2.1 illustrates representative input-target mappings. These short examples show the structure of each task; the experimental sequence lengths are specified in Chapter 3.

Length generalisation should also be distinguished from long-context modelling. Architectures such as Transformer-XL extend the amount of context that can be processed or reused [6], while benchmarks such as Long Range Arena compare efficiency and performance on tasks containing long-range dependencies [26]. These are important related problems, but a model may technically accept a long input without applying a rule learned at shorter lengths. The present study focuses on this latter behavioural question.

2.4 Evaluating Length-Generalisation Failure

A valid extrapolation claim requires evidence that the model first learned the training task. If training-length performance is poor, failure on a longer input cannot be attributed specifically to the length shift. The baseline-eligibility procedure used in this dissertation therefore separates configurations that reliably learn the training task from those exhibiting underfitting or seed-dependent success.

Sequence-level exact-match accuracy is a strict measure of whether the complete predicted output is correct. It is appropriate for deterministic tasks, but it becomes increasingly demanding as the output grows: a single incorrect token makes the complete sequence incorrect. Token accuracy and token-error percentage are therefore also required. They reveal whether zero exact-match accuracy represents almost-correct sequences or widespread token-level failure.

Results from a single random seed can also be misleading. Neural model performance may vary with parameter initialisation, data ordering, and stochastic optimisation, and reporting practices should make this variation visible [20]. The main benchmark therefore uses three seeds and distinguishes consistent training fit from success observed only in individual runs.

Aggregate accuracy still does not show where errors occur. Position-wise profiles can reveal whether failures concentrate near sequence boundaries or are distributed throughout the output. Task-specific groupings provide further tests: incoming-carry conditions are relevant to addition, while dependency distance is relevant to reverse. These analyses support a more precise description of failure than the statement that a model simply "does not generalise."

Attention statistics provide an additional, but limited, source of evidence. Entropy, attention distance, and local-attention ratios can describe how attention distributions change with sequence length. They cannot, by themselves, establish the causal mechanism responsible for a prediction. Attention patterns should therefore be interpreted as behavioural diagnostics rather than complete explanations of model reasoning [10].

2.5 Training Interventions

Several training strategies provide plausible hypotheses for improving length generalisation. Mixed-length training exposes the model to more than one sequence length and may reduce dependence on a single fixed training length. It does not, however, guarantee extrapolation beyond the largest length represented during training.

Curriculum learning presents examples in a structured progression, commonly from easier to more difficult instances [4]. In a length curriculum, progressively increasing sequence length may help optimisation by allowing shorter-task solutions to be refined at later stages. A curriculum that fails to preserve training performance cannot be interpreted as a successful extrapolation intervention.

Randomised positional exposure aims at a different possible mechanism. Randomised positional encodings have been shown to improve Transformer length generalisation in some experimental settings by exposing the model to a wider set of positions during training [21]. The randomised padded condition in this dissertation tests a related hypothesis: varying the absolute locations occupied by meaningful tokens may reduce reliance on the fixed positions encountered in the original baseline.

These interventions are not treated as interchangeable solutions. Each tests a particular hypothesis: mixed-length training changes length exposure, curriculum learning changes the optimisation schedule, and randomised padding changes absolute-position exposure. Their results must therefore be compared with matched baselines and interpreted according to whether they preserve training-length performance and improve unseen-length performance.

2.6 Real-World Long-Sequence Context

Transformer encoders have been applied to multivariate time-series representation learning and classification, including datasets with relatively few labelled examples [30]. Specialised Transformer designs have also been developed for long or irregular temporal inputs [16], while the UEA archive provides established multivariate classification benchmarks [2].

The present extension uses SelfRegulationSCP2 only as an exploratory contrast. It predicts one class label rather than a complete output sequence, so its classification accuracy is not directly comparable with synthetic exact-match accuracy. It is therefore not presented as a state-of-the-art time-series benchmark or as validation of the complete synthetic taxonomy.

2.7 Research Gap

Existing research identifies positional representation, training distribution, inductive bias, and computational structure as interacting determinants of Transformer extrapolation [19, 11, 21, 17, 29]. Recent arithmetic studies further show that task-aligned representations or explicit intermediate computation can improve extrapolation in particular settings [1, 13, 15, 9]. These findings do not establish one positional encoding or training intervention that transfers reliably across all tasks.

A remaining difficulty is interpreting poor long-length accuracy. An aggregate result cannot distinguish a model that underfits the training task from one that learns it reliably before collapsing outside the training distribution. It may also conceal seed instability and differences in where errors occur. Two configurations with the same zero exact-match accuracy may have substantially different token-error percentages, positional profiles, or task-specific error structures and therefore require different interpretations.

Mitigation experiments introduce a related problem. Mixed-length exposure, curriculum learning, and randomised positional exposure test different hypotheses about failure. An intervention cannot be interpreted as improving length generalisation if it does not preserve training-length performance, and its result should be compared with the corresponding original baseline.

This dissertation addresses this evaluation gap rather than proposing a new Transformer architecture. It combines multi-seed baseline eligibility, length-wise degradation curves, exact-match and token-error measures, position-wise and task-specific diagnostics, descriptive attention statistics, and matched baseline–intervention comparisons. The resulting framework distinguishes training underfitting, seed instability, immediate extrapolation collapse, spatial error structure, and no observed improvement under tested mitigation. It therefore describes not only whether performance fails at longer lengths, but what form that failure takes and which intervention hypothesis was evaluated.

Chapter 3

Methodology and Experimental Design

This chapter describes the methodology used to investigate length generalisation in compact Transformer models. The experiments follow a controlled train-short/test-long protocol: a model is trained using sequences of a fixed length and subsequently evaluated on progressively longer instances of the same task.

The methodology was designed around three requirements. First, a configuration had to learn the training-length task reliably before its longer-length performance could be interpreted. Second, the main baseline conclusions had to be supported across multiple random seeds. Third, evaluation had to describe not only whether accuracy decreased, but also when failure began and where errors occurred within the output.

The chapter specifies the synthetic tasks, evaluation lengths, model architecture, positional encoding implementations, training procedure, metrics, diagnostic analyses, mitigation experiments, failure-classification procedure, and exploratory real-world extension.

3.1 Experimental Design

The main benchmark crossed three synthetic tasks - Addition, Delayed Copy, and Reverse with five positional encoding conditions and three random seeds. The positional conditions were learned absolute embeddings, sinusoidal encoding, no explicit positional encoding (NoPE), ALiBi, and RoPE. This produced

$$3 \text{ tasks} \times 5 \text{ positional encodings} \times 3 \text{ seeds} = 45$$

independently trained synthetic baseline models. The complete experimental matrix, including the synthetic, real-world, and targeted mitigation conditions, is reported in Appendix B.1. Each checkpoint was evaluated at every test length configured for its task.

The experiment proceeded in stages. Training-length results were first used to determine whether a task–encoding configuration provided an eligible baseline. Eligible configurations were then examined across longer lengths using exact-match accuracy, token error, position-wise error profiles, task-specific diagnostics, and attention statistics. Finally, targeted training interventions were compared with their corresponding original baselines.

This ordering prevents different experimental outcomes from being conflated. A configuration that never learns the training task represents underfitting. A configuration that succeeds for some seeds but not others represents seed instability. Extrapolation collapse is considered only after stable training-length performance has been established.

A separate experiment crossed the SelfRegulationSCP2 time-series dataset with the same five positional encoding conditions and three seeds, producing 15 additional models. It was analysed separately because it predicts one class label rather than a deterministic target sequence.

Task	L_{train}	L_{test}	Vocabulary and encoded length
Addition	16	16, 32, 64, 128, 256, 512, 1024	10 digits, plus, and padding; encoded length $2L+1$
Delayed Copy	128	128, 256, 512, 1024	100 content symbols, separator, and padding; encoded length $2L+1$
Reverse	128	128, 256, 512, 1024	100 content symbols; encoded length L

Table 3.1: Synthetic-task training and evaluation schedule. Lengths denote digits per operand for Addition, content tokens for Delayed Copy, and complete sequence length for Reverse.

3.2 Synthetic Tasks and Length Protocol

The conceptual differences between the synthetic tasks were introduced in Chapter 2 and illustrated in Figure 2.1. This section specifies their experimental implementation.

For each task, the nominal length L has a different meaning. In Addition, L is the number of digits in each operand. Two L-digit operands are separated by a plus token, producing an input of length $2L+1$. Leading zeros are excluded so that both operands have the intended length. The target contains the calculated sum right-aligned within a sequence of the same encoded length, with preceding positions filled by a padding token.

In Delayed Copy, L is the number of content tokens that must be reproduced. The input contains L randomly sampled content tokens, a separator, and L padding tokens. The target contains $L+1$ padding tokens followed by the original content sequence. Its encoded input and target lengths are therefore also $2L+1$.

In Reverse, L is the complete input length. The input consists of L randomly sampled content tokens, and the target contains the same tokens in the opposite order. No padding or separator token is required.

A separate held-out test set was generated at every evaluation length. The data-generating rule and vocabulary remained unchanged, so the experimental shift concerned sequence length rather than task identity. Addition was first evaluated beyond training at $L = 32$, while Delayed Copy and Reverse were first evaluated beyond training at $L = 256$.

The protocol follows the train-short/test-long structure used in prior work on positional extrapolation and algorithmic length generalisation [19, 11, 31, 27]. However, this study adds an explicit multi-seed eligibility stage before interpreting longer-length results.

Associative Recall was implemented during development but excluded from the main matrix because the model did not learn it reliably at the training length. Including it would have made it impossible to distinguish ordinary underfitting from failure caused specifically by the length shift.

3.3 Model Architecture

The synthetic experiments used a compact encoder-style Transformer implemented in PyTorch [18]. The model was trained from scratch and consisted of a token-embedding layer, one positional treatment, two Transformer blocks, and a linear output projection applied independently at every sequence position.

Each Transformer block contained four-head self-attention followed by a position-wise feedforward network. Residual connections and dropout were applied around both sublayers, with layer normalisation performed after each residual addition. The feedforward network used a ReLU activation.

The model used full bidirectional self-attention without a causal mask. This matches the non-autoregressive formulation of the controlled tasks: the complete input was supplied simultaneously, and the model predicted a target token at every position.

The maximum supported encoded length was 2049 for Addition and Delayed Copy, corresponding to $2(1024) + 1$, and 1024 for Reverse. The architecture was otherwise held constant across positional conditions. Its compact size made the multi-task, multi-seed experiment feasible while retaining the principal components of the Transformer architecture [28].

Architecture component	Configuration
Token-embedding dimension	128
Transformer blocks	2
Attention heads	4
Dimension per attention head	32
Feedforward hidden dimension	256
Feedforward activation	ReLU
Dropout	0.1
Normalisation	Post-residual layer normalisation
Attention visibility	Full bidirectional attention
Output	Vocabulary logits at every position

Table 3.2: Transformer architecture used in the synthetic experiments.

Condition	Application point	Implementation
Learned	Input representation	A trainable vector was added for every supported absolute position.
Sinusoidal	Input representation	Fixed sine and cosine vectors were added using the original Transformer frequency schedule.
NoPE	No explicit addition	Token embeddings entered the first Transformer block without an added positional signal.
ALiBi	Attention logits	A head-specific negative bias proportional to absolute query–key distance was added before the softmax.
RoPE	Queries and keys	Position-dependent rotations were applied to paired query and key dimensions before their dot product.

Table 3.3: Implementation of the five positional encoding conditions.

3.4 Positional Encoding Conditions

The theoretical motivations for the five positional encoding methods were reviewed in Chapter 2. Their experimental implementations are summarised in Table 3.3. Only the positional treatment changed; the remaining architecture and baseline training procedure were held fixed.

For the learned condition, an embedding table was allocated for the complete supported evaluation range. During baseline training, only entries corresponding to positions occurring at the training length received gradient updates. Embeddings used exclusively at unseen positions therefore remained at their initial values. This property is important when interpreting the learned absolute-position baseline.

Sinusoidal encodings were precomputed across the complete supported range using the original Transformer construction [28]. No new positional parameters were therefore required at longer evaluation lengths.

The NoPE condition applied no explicit positional transformation. It retained the same full bidirectional attention and feedforward operations as the other conditions. It was included because previous work has shown that the behaviour of NoPE is dependent on task and architecture [11].

ALiBi was adapted to the bidirectional model by adding

$$B_{ij}^{(h)} = -m_h |i - j|$$

to the attention score between positions i and j in head h, where m_h is the fixed head-specific slope. The absolute distance makes this a symmetric adaptation of the original causal ALiBi formulation [19].

RoPE was applied to queries and keys, but not values, using an even per-head dimension of 32 and the standard frequency base of 10 000 [24].

3.5 Training and Evaluation

For each synthetic configuration, 10,000 training examples, 1,000 validation examples, and 1,000 test examples at every evaluation length were generated. The validation split was retained for pipeline checking but was not used for early stopping or checkpoint selection.

Training component	**Configuration**
Training examples	10,000 per baseline run
Validation examples	1,000 at the training length
Test examples	1,000 at each evaluation length
Training epochs	10
Optimiser	AdamW
Learning rate	10^{-3}
Weight decay	0.01
Checkpoint selection	Final epoch
Addition batch size	8
Delayed Copy batch size	1
Reverse batch size	1
Seeds	42, 123, and 2024

Table 3.4: Training and evaluation configuration for the synthetic baselines.

Each baseline model was trained for 10 epochs using token-level cross-entropy loss and AdamW [14]. The learning rate was 10^{-3}, and the PyTorch default weight decay of 0.01 was retained. No learning-rate scheduler, gradient clipping, or early stopping was applied. The checkpoint from the final epoch was used for every evaluation.

Padding targets were excluded from the loss for Addition and Delayed Copy. Reverse has no padding target, so every position contributed to its loss.

The recorded seed controlled Python, NumPy, and PyTorch randomness, including synthetic data generation, model initialisation, minibatch order, and CUDA randomness when available. The three repetitions therefore measure the combined sensitivity of the pipeline to data sampling and stochastic training.

After training, each checkpoint was loaded in evaluation mode and tested separately on every held-out length. Dropout was disabled, and the predicted token at each position was the highest-logit vocabulary item. Test data were not used for model selection or subsequent optimisation.

3.5.1 Evaluation metrics

Let $y_{n,j}$ and $\hat{y}_{n,j}$ denote the target and predicted tokens at position j in example n. Let $m_{n,j} = 1$ for an evaluated target position and zero for padding. Token accuracy at test length L is

$$A_{\text{tok}}(L) = 100 \frac{\sum_{n,j} m_{n,j} \mathbb{I}[\hat{y}_{n,j} = y_{n,j}]}{\sum_{n,j} m_{n,j}}. \tag{3.1}$$

Exact-match accuracy is

$$A_{\text{EM}}(L) = \frac{100}{N} \sum_{n=1}^{N} \mathbb{I}\left[\hat{y}_{n,j} = y_{n,j} \text{ for every evaluated } j\right]. \tag{3.2}$$

Exact match is the primary synthetic metric because each task requires a complete structured output. Token accuracy provides complementary evidence about partial performance. Both were also expressed as error percentages:

$$E_{\text{tok}}(L) = 100 - A_{\text{tok}}(L), \qquad E_{\text{EM}}(L) = 100 - A_{\text{EM}}(L).$$

For context, uniform prediction over the active target vocabulary corresponds to 90% token error for Addition and 99% for Delayed Copy and Reverse. These are token-level references, not exact-match thresholds.

For each task, encoding, and length, the mean, sample standard deviation, and standard error were calculated across the three seeds. Standard deviation describes observed variation between runs, whereas standard error describes uncertainty in the estimated mean. Figure captions state explicitly which quantity is displayed.

3.5.2 Baseline eligibility and collapse

A seed was considered to have fitted its training task when

$$A_{\mathrm{EM}}(L_{\mathrm{train}}) \geq 90\%.$$

A task-encoding configuration was classified as eligible only when all three seeds met this threshold. If only one or two seeds met it, the configuration was classified as seed unstable. If no seed met it, the configuration was not eligible for the principal extrapolation analysis.

The 90% threshold is an operational criterion rather than a claim of perfect learning. Its purpose is to prevent training underfitting or seed-specific success from being misreported as length-generalisation failure.

For an eligible configuration, the first collapse length was defined as

$$L_{\mathrm{collapse}} = \min\left\{L > L_{\mathrm{train}} : \bar{A}_{\mathrm{EM}}(L) < 10\%\right\}, \tag{3.3}$$

where $\bar{A}_{\mathrm{EM}}(L)$ is the three-seed mean. Falling below this threshold at the first unseen length was described as immediate extrapolation collapse. The 10% value is a consistent reporting threshold; it is not the theoretical chance probability of predicting a complete sequence correctly and does not identify the internal cause of failure.

3.6 Experimental Controls and Verification

Several controls were used to reduce alternative explanations for observed differences between positional encoding conditions. Within each task and random seed, the synthetic generator used the same task parameters and seed for every positional encoding. Consequently, positional encoding conditions with the same task and seed were trained and evaluated using identically generated examples. Differences between those conditions cannot therefore be attributed to different sampled datasets.

Within each task, the Transformer architecture, optimisation settings, training duration, vocabulary, and evaluation schedule were also held fixed across positional encodings. Batch size differed between tasks because of their different encoded lengths and memory requirements, but it did not vary between positional encodings within the main baseline comparison. The cross-task analysis therefore compares qualitative failure behaviour rather than claiming that raw task accuracies constitute a capacity-controlled ranking of task difficulty.

The baseline and extrapolation analyses used all three seeds. Detailed position-wise, carry-condition, dependency-distance, and attention analyses used representative seed-42 checkpoints from the five eligible task-encoding configurations. Position-wise diagnostics used all 1,000 test examples at each evaluated length, whereas attention statistics used the first five test batches. Mitigation experiments used targeted seed-42 interventions and their corresponding seed-42 baselines. The real-world extension retained all three seeds.

This distinction separates multi-seed conclusions from representative diagnostic case studies. The seed-42 diagnostics characterise how selected eligible models failed; they do not establish the prevalence of each diagnostic pattern across seeds.

Training, validation, and test splits were generated and stored separately. The validation split was not used for early stopping or model selection, and test results did not influence the saved checkpoint. The same final-epoch checkpoint was loaded for every evaluation length, ensuring that changes in performance resulted from the evaluation input rather than additional training.

For Addition and Delayed Copy, padding targets were masked consistently in both loss and evaluation. This prevents the large number of padding positions from artificially increasing token accuracy or causing a sequence to be classified as incorrect. Position-wise diagnostic evaluations independently recomputed aggregate token and exact-match accuracy and checked them against the corresponding evaluation CSV before producing error profiles.

The complete control matrix, including the potential confounds, corresponding experimental controls, and interpretive consequences, is provided in Appendix B.3.

3.7 Failure Diagnostics

The principal baseline curves established whether and when exact-match performance collapsed. Additional diagnostics were applied to the five eligible task-encoding configurations: Addition with learned and sinusoidal encodings, Delayed Copy with sinusoidal and RoPE, and Reverse with learned positional embeddings.

3.7.1 Position-wise and task-specific errors

Position-wise diagnostics used the representative seed-42 checkpoints and all 1,000 held-out examples at each evaluation length. Their aggregate token and exact-match accuracies were checked against independently generated evaluation CSVs before positional errors were analysed.

Padding targets were removed, and the remaining target positions in each example were divided into ten equal relative-position bins. The error percentage for bin b was calculated as

$$E_b = 100 \frac{\text{incorrect evaluated tokens in bin } b}{\text{evaluated tokens in bin } b}. \tag{3.4}$$

Using relative rather than absolute bins allows outputs of different lengths to be compared. The first bin represents the beginning of the valid target, and the tenth represents its end.

Addition target digits were additionally grouped by whether their calculation received an incoming carry from the less significant column. An optional leading carry digit was recorded separately. This tested whether failure was concentrated on the recurrent dependency required for carry propagation.

For Reverse, each output position was associated with its required input source position. Their absolute separation was normalised by the maximum possible distance and divided into ten bins. This tested whether errors were more frequent when the required source token was farther from its output position.

These diagnostics describe the seed-42 checkpoints and are not presented as multi-seed estimates. Multi-seed eligibility and seed stability are reported separately.

3.7.2 Attention diagnostics

Attention was examined at the training length, first unseen length, and longest test length for the same five eligible configurations. Three statistics were calculated:

- normalised attention entropy, measuring how concentrated or diffuse the attention distribution was relative to its maximum entropy;
- average attention distance, measuring the expected absolute separation between query and attended key positions; and
- local attention ratio, measuring the proportion of attention assigned within ten positions of each query.

Statistics were averaged over query positions, attention heads, and both Transformer layers. Attention was collected from the first five test batches. This corresponded to at most 40 Addition examples and five Delayed Copy or Reverse examples because their baseline batch sizes differed.

The attention results are therefore representative supporting diagnostics, not a complete multi-seed mechanistic analysis. Attention weights are not necessarily faithful explanations of a model's predictions [10]. Furthermore, unnormalised distance has a larger possible range at longer lengths, while a fixed local window represents a smaller fraction of a longer sequence. These statistics were consequently used to describe accompanying changes rather than assign a causal explanation to failure.

3.8 Mitigation Experiments

The mitigation study used seed 42 and compared each intervention with the corresponding seed-42 original baseline. It was designed as a set of targeted follow-up experiments rather than a second complete multi-seed matrix.

Three interventions were evaluated. Mixed-length training tested whether broader exposure below and including the original training length reduced dependence on a single length. Curriculum learning tested whether increasing length gradually overcame an optimisation difficulty [4]. Randomised padded training tested whether variation in the positions occupied by the separator and target reduced dependence on fixed absolute locations. This final intervention was motivated by research on randomised positional exposure [21], but it was a narrower experiment rather than a direct replication.

Intervention	Conditions	Training design	Hypothesis
Mixed length	Addition-Learned; Delayed Copy-Sinusoidal	Examples drawn from lengths below and including the baseline training length	Broader length exposure may reduce reliance on one fixed length.
Curriculum	Delayed Copy-Sinusoidal; Delayed Copy-RoPE; Reverse-Learned	Sequential stages at lengths 32, 64, 96, and 128	Gradual increases may reduce an optimisation barrier at length 128.
Randomised padded	Delayed Copy-Sinusoidal; Delayed Copy-RoPE	Lengths 32, 64, 96, and 128 shuffled and right-padded to a common maximum	Varying separator and target positions may reduce fixed-position dependence.
Single-length control	Delayed Copy-Sinusoidal	Length-128 data trained through the mitigation pipeline	Separates the intervention from changes introduced by the alternative pipeline and batch size.

Table 3.5: Mitigation conditions and the specific hypotheses they test.

The first Addition mixed-length variant distributed 10,000 examples across operand lengths 4, 8, 12, and 16 and trained for 10 epochs. A strengthened variant used 40,000 examples and 20 epochs to test whether the first condition was limited by training volume.

The Delayed Copy mixed-length and randomised-padded conditions used 40,000 examples distributed across lengths 32, 64, 96, and 128. Mixed-length training processed separate fixed-length datasets. Randomised-padded training combined and shuffled the examples before right-padding them to the maximum training length. Examples were not inserted at random offsets; the content, separator, and output positions varied because task length varied.

Curriculum models were trained sequentially at lengths 32, 64, 96, and 128. Each stage used 10,000 examples and five epochs, and the final checkpoint from one stage initialised the next. Only the final length-128 checkpoint was evaluated over the complete test schedule.

The mitigation pipeline used batch size eight, whereas the original Delayed Copy and Reverse baselines used batch size one. The single-length Delayed Copy control therefore tested whether the alternative pipeline could reproduce the original training-length result without changing length exposure.

An intervention had to retain

$$A_{\mathrm{EM}}(L_{\mathrm{train}}) \geq 90\%$$

before its extrapolation result was interpreted. Otherwise, the comparison was classified as inconclusive due to training underfitting. For an eligible intervention, improvement at length L was measured relative to its matched baseline as

$$\Delta A_{\mathrm{EM}}(L) = A_{\mathrm{EM}}^{\mathrm{intervention}}(L) - A_{\mathrm{EM}}^{\mathrm{baseline}}(L). \tag{3.5}$$

The wording *no observed improvement under the tested mitigation* is used when an eligible intervention did not improve performance at the evaluated unseen lengths. This does not imply that the task is resistant to all possible mitigation.

The Reverse curriculum used a 102-symbol vocabulary, whereas its original baseline used 100 symbols. It is therefore not a fully matched baseline-intervention comparison and is treated only as an exploratory curriculum sensitivity test.

3.9 Failure Taxonomy Construction

The observed results were organised using five sequential questions:

1. Did the configuration learn the training-length task?
2. Was training success stable across all three seeds?
3. At which unseen length did severe degradation first occur?
4. How were errors distributed across positions and task-specific conditions?
5. Did an intervention retain training fit and improve unseen-length performance relative to its matched baseline?

These questions distinguish training-length underfitting, seed instability, immediate extrapolation collapse, spatial error structure, and no observed improvement under tested mitigation. The categories are descriptive and need not be mutually exclusive. For example, an eligible baseline can display immediate collapse together with an approximately uniform position-wise error profile.

Attention changes were retained as supporting observations but were not used alone to assign a category. Likewise, a mitigation that underfit the training task was classified as inconclusive rather than as evidence that the underlying failure could not be mitigated. The resulting taxonomy is therefore an evidence-organising framework, not a claim that every category corresponds to a unique internal mechanism.

3.10 Exploratory Real-World Extension

The synthetic benchmark was supplemented by an exploratory experiment usingSelfRegulationSCP2 from the UEA Multivariate Time Series Classification Archive [2, 30]. This binary classification dataset contains seven-channel EEG recordings, with 200 training and 180 test examples of 1152 time points. Each channel was standardised using statistics calculated only from the training split.

Models were trained on prefixes of length 256 and evaluated using prefixes of length 256, 512, 1024, and 1152. The same two-block Transformer, five positional conditions, and three seeds were used, with a linear input projection, mean pooling, and a two-class output layer. Mean accuracy and standard error were reported across seeds. Because longer prefixes may add useful, irrelevant, or noisy observations, this classification experiment is an exploratory contrast rather than a direct equivalent of deterministic synthetic length extrapolation. Consequently, the synthetic eligibility and collapse thresholds were not applied.

3.11 Implementation and Reproducibility

The project was implemented in Python using PyTorch. Synthetic task generation, model definitions, training, evaluation, diagnostics, and figure generation were implemented as separate modules. Experimental settings were stored in YAML configuration files, while raw seed-level results were retained separately from derived summaries and report figures.

Local macOS execution was used for development, validation, aggregation, and figure generation. GPU-intensive training and diagnostic evaluation were performed on an Amazon Web Services G5-family instance using the same Git repository and configuration interface.

Configurations preserve the task, length schedule, model settings, optimisation settings, seed, checkpoint location, and output location. Together with the saved checkpoints, raw CSV files, and analysis scripts, these records support reconstruction of the principal experimental workflow. Exact dependency versions were not pinned, so byte-for-byte environment reproduction is not guaranteed. Detailed repository traceability is provided in Appendix B.4.

The public software repository and principal reproduction entry points are documented in Appendix C.

Chapter 4

Experimental Results and Failure Analysis

4.1 Synthetic Baseline Eligibility and Extrapolation

This chapter reports the main synthetic benchmark results. It first determines which task-encoding configurations reliably learned their training task across all three seeds. It then examines whether those eligible baselines transferred to longer sequences and uses token error to distinguish complete sequence failure from partial token-level performance.

The principal result is consistent across the three tasks: every eligible baseline fell below the 10% exact-match threshold at the first unseen length. However, the token-level form of this collapse differed between Addition and the two sequence-transformation tasks.

4.1.1 Training Fit and Baseline Eligibility

Figure 4.1 reports training-length exact-match accuracy for all 45 synthetic baseline runs. Each marker represents one seed, the short horizontal line represents the three-seed mean, and the dashed line marks the 90% training-fit threshold. The figure displays individual runs rather than using an error bar, making seed instability directly visible.

Five of the 15 task–encoding configurations were eligible:

- Addition with learned positional embeddings;
- Addition with sinusoidal encoding;
- Delayed Copy with sinusoidal encoding;
- Delayed Copy with RoPE; and
- Reverse with learned positional embeddings.

Addition with learned embeddings achieved $92.00 \pm 1.84\%$ training-length exact match, while sinusoidal encoding achieved $93.70 \pm 0.95\%$. The uncertainty values are sample standard deviations across the three seeds. All three seeds exceeded the eligibility threshold in both conditions.

Delayed Copy with sinusoidal encoding achieved $99.93 \pm 0.12\%$, and RoPE achieved $100.00 \pm 0.00\%$. Reverse with learned positional embeddings also achieved $100.00 \pm 0.00\%$. These three conditions therefore provided particularly stable training-length baselines.

Two additional configurations were seed unstable. Delayed Copy with learned embeddings succeeded only for seed 123, which achieved 99.4% exact match; seeds 42 and 2024 achieved 0%. Its pooled mean of 33.13% therefore obscures a successful and two unsuccessful runs. Reverse with sinusoidal encoding succeeded for seeds 42 and 123, which achieved 100.0% and 99.8%, respectively, while seed 2024 achieved 0%. Neither configuration was treated as an eligible pooled baseline.

The remaining eight configurations were not eligible. NoPE and ALiBi achieved 0% training-length exact match on all three tasks. Addition with RoPE partially fitted the training task, achieving $47.30 \pm 3.60\%$, but no seed reached the 90% threshold. Reverse with RoPE also underfit. These outcomes cannot be interpreted as extrapolation collapse because the corresponding training-length task was not first learned reliably.

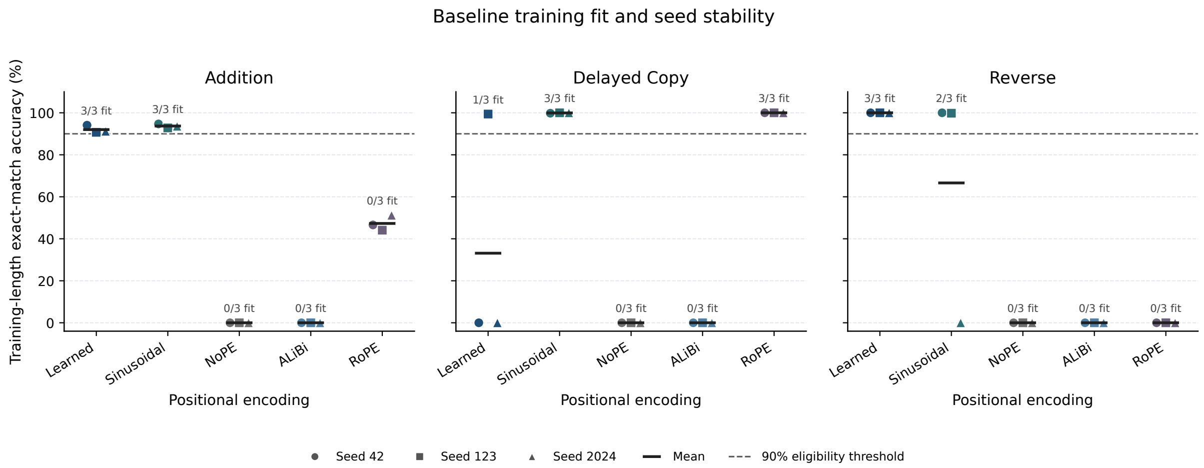

Figure 4.1: Training-length exact-match accuracy for every task, positional encoding, and random seed. Circles, squares, and triangles denote seeds 42, 123, and 2024, respectively; horizontal bars show the three-seed mean. Labels report the number of seeds reaching the 90% eligibility threshold.

Task	**Encoding**	**Exact at train, mean $\pm$ SD**	**First unseen length**	**Token error at first unseen, mean $\pm$ SD**
Addition	Learned	92.00 ± 1.84	32	87.74 ± 0.23
Addition	Sinusoidal	93.70 ± 0.95	32	87.16 ± 0.48
Delayed Copy	Sinusoidal	99.93 ± 0.12	256	99.00 ± 0.01
Delayed Copy	RoPE	100.00 ± 0.00	256	98.98 ± 0.02
Reverse	Learned	100.00 ± 0.00	256	98.75 ± 0.03

Table 4.1: Training-length exact-match accuracy and first-unseen token-error percentage for the five eligible baselines. Every first-unseen exact match accuracy was 0%. Values are the mean and sample standard deviation across three seeds.

4.1.2 Extrapolation of Eligible Baselines

Table 4.1 summarises the complete set of eligible baselines. Exact match fell to 0% at the first unseen length for all five configurations and remained at 0% at every subsequent length. Consequently, each configuration crossed the operational collapse threshold immediately beyond its training range.

Addition

The two eligible Addition baselines were trained at operand length 16 and collapsed at length 32. Although complete-sequence exact match fell to zero, token error was 87.74% for learned embeddings and 87.16% for sinusoidal encoding. The models therefore retained approximately 12–13% token accuracy at the first unseen length.

Figure 4.2 shows that both eligible Addition baselines crossed the collapse threshold at the first unseen length. The agreement between the individual seed curves confirms that this result is not produced by averaging incompatible runs.

As length increased further, token error approached the 90% uniform ten-digit reference. At length 1024, learned and sinusoidal token errors were 89.93% and 89.98%, respectively. The initial failure was therefore already close to uniform digit prediction and became still closer to that reference at the longest length.

This result differs from a prediction that was mostly correct but contained one or two incorrect digits. Exact match was zero because token-level performance had deteriorated across a substantial portion of the output.

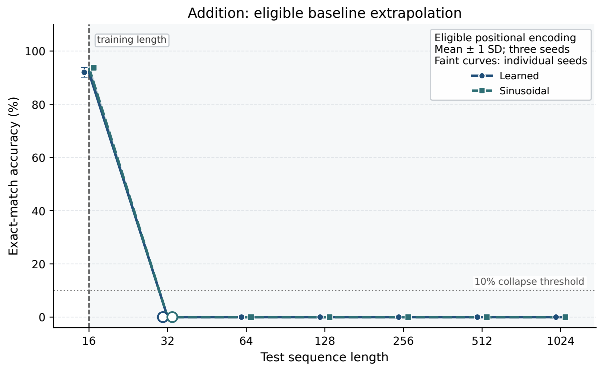

Figure 4.2: Exact-match accuracy across test lengths for the eligible Addition baselines. Bold curves show the three-seed mean, shaded regions show ±1 sample standard deviation, and faint curves show individual seeds. The vertical dashed line marks the training length, the horizontal dotted line marks the 10% collapse threshold, and the enlarged open marker identifies the first collapse length.

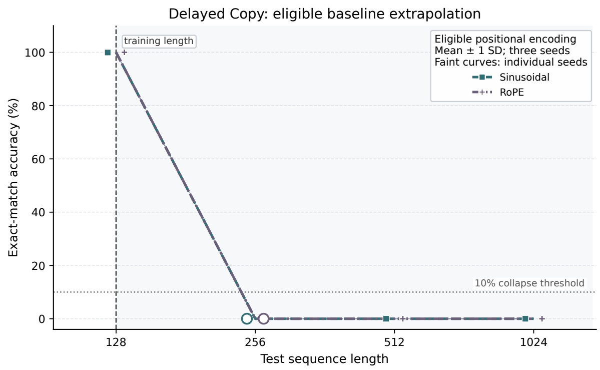

Figure 4.3: Exact-match accuracy across test lengths for the eligible Delayed Copy baselines. Bold curves show the three-seed mean, shaded regions show ±1 sample standard deviation, and faint curves show individual seeds. The vertical dashed line marks the training length, the horizontal dotted line marks the 10% collapse threshold, and enlarged open markers identify the first collapse length.

Delayed Copy

The eligible Delayed Copy baselines were trained at length 128 and collapsed at length 256. Sinusoidal encoding fell from 99.93% mean training-length exact match to 0%, while RoPE fell from 100% to 0%.

Figure 4.3 shows that the sinusoidal and RoPE curves are nearly coincident. Both configurations retained stable training-length performance across seeds and then collapsed at the same first unseen length.

Their first-unseen token errors were 99.00% and 98.98%, respectively, approximately equal to the 99% uniform reference for 100 content symbols. Token error remained near 99% at lengths 512 and 1024. The failure therefore affected the copied content broadly rather than producing outputs that were mostly correct but failed strict sequence-level exact match.

Reverse

Reverse with learned positional embeddings achieved 100% exact match for every seed at length 128. At length 256, mean exact match fell to 0% and token error reached 98.75%. It increased to 98.84% at length 512 and 98.92% at length 1024, approaching the 99% uniform reference.

Figure 4.4 shows that the result is identical across the three learned-encoding seeds: perfect performance at the training length is followed by complete exact-match collapse when the nominal sequence length doubles.

The learned absolute-position model therefore fitted the length-128 reversal precisely but did not transfer that mapping when the sequence length doubled. As with Delayed Copy, the zero exact-match result reflected widespread token-level failure rather than a small number of misplaced tokens.

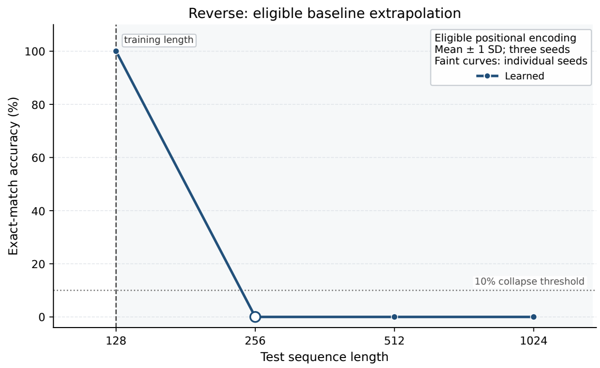

Figure 4.4: Exact match accuracy across test lengths for the eligible Reverse baseline. The bold curve shows the three-seed mean, the shaded region shows ±1 sample standard deviation, and faint curves show individual seeds. The vertical dashed line marks the training length, the horizontal dotted line marks the 10% collapse threshold, and the enlarged open marker identifies the first collapse length.

4.1.3 Comparison Across Positional Encodings

Positional encoding strongly affected whether a training task was learned, but no encoding produced an eligible baseline across all three tasks. Learned embeddings were eligible for Addition and Reverse but seed unstable for Delayed Copy. Sinusoidal encoding was eligible for Addition and Delayed Copy but seed unstable for Reverse. RoPE was eligible only for Delayed Copy.

NoPE and ALiBi did not produce an eligible baseline under the selected architecture and optimisation procedure. Their longer-length results cannot therefore show whether these mechanisms would extrapolate after successful training. The defensible conclusion is limited to training underfitting in this experimental setting.

Similarly, RoPE's success on Delayed Copy does not establish a general advantage across tasks. It achieved only partial training fit on Addition and underfit Reverse. Conversely, learned absolute positions solved Reverse at the training length but failed immediately at the first unseen length. The results therefore do not support a task-independent ranking of positional encoding methods.

4.1.4 Seed Stability and Interpretation

The multi-seed analysis changes the interpretation of several configurations. A pooled mean alone would describe Delayed Copy with learned embeddings as a poor model with 33.13% training exact match. The individual runs instead show a bimodal outcome: one seed learned the task almost perfectly, while two failed completely. Reverse with sinusoidal encoding displayed the same type of instability with two successful seeds and one failed seed.

This justifies requiring every seed to meet the training-fit threshold before pooling a configuration as an extrapolation baseline. The criterion is conservative, but it prevents isolated successful runs from being presented as stable properties of an encoding.

The eligible configurations were stable at the training length and also consistent in their first-unseen collapse: every seed reached 0% exact match. The principal conclusion is therefore not produced by averaging incompatible seed behaviours. It is supported by both the multi-seed mean and the individual eligible runs.

4.2 Failure Behaviour Diagnostics

Section 4.1 established that all five eligible task-encoding baselines fell below the 10% exact-match threshold at the first evaluated unseen length. This chapter examines how those failures were distributed and whether the tested interventions changed the observed behaviour. The analysis combines position-wise token errors, task-specific diagnostics, descriptive attention statistics, and matched baseline-mitigation comparisons.

The detailed diagnostics use representative seed-42 checkpoints. They therefore characterise the form of failure in selected trained models rather than its prevalence across random initialisations. Multi-seed baseline eligibility and seed stability were established separately in Section 4.1.

4.2.1 Where Failure Occurs

Exact-match accuracy determines whether an entire prediction is correct, but it cannot show where errors occur once sequence-level accuracy reaches zero. The target positions were therefore divided into relative-position bins, and token-error rates were calculated within each bin. The analysis was restricted to the five task-encoding configurations that satisfied the multi-seed baseline-eligibility criterion.

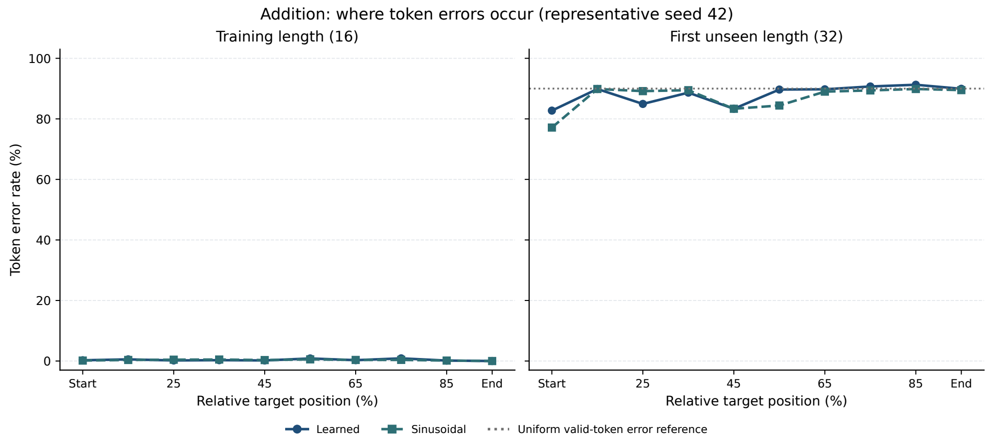

Figure 4.5: Position-wise token-error rates for eligible Addition configurations at the training length and first evaluated unseen length. Target positions are grouped into relative-position bins. The dotted line is the 90% error expected under uniform prediction over the ten valid digits. Results use representative seed-42 checkpoints.

Figure 4.5 shows that the Addition models had low error throughout the output at the training length. At length 32, errors occurred across the entire target, although later positions were somewhat less accurate. The learned model's error increased from 82.7% in the start region to 89.9% near the end. The sinusoidal model showed a larger increase, from 77.1% to 89.5%.

Task	Encoding	Overall	Start	Middle	End	Range
			Token error (%)			(pp)
Addition	Learned	88.0	82.7	86.5	89.9	8.5
Addition	Sinusoidal	86.9	77.1	83.9	89.5	12.7
Delayed Copy	Sinusoidal	99.0	98.9	99.0	99.1	0.2
Delayed Copy	RoPE	99.0	99.0	98.9	99.1	0.2
Reverse	Learned	98.8	98.9	98.8	98.3	0.7

Table 4.2: Token-error distribution at the first evaluated unseen length. Start, middle, and end correspond to grouped relative-position regions. Range is the difference between the largest and smallest regional error rate in percentage points. Results use representative seed-42 checkpoints.

Table 4.2 compares this behaviour with Delayed Copy and Reverse. Addition retained a modest positional gradient at the first unseen length. In contrast, Delayed Copy and Reverse produced error rates close to the 99% uniform valid-token reference throughout the output, with less than one percentage point of variation between their start, middle, and end regions.

These results distinguish two forms of failure that are hidden by the common 0% exact-match result. Addition retained limited position-dependent structure, particularly near the beginning of the output. Delayed Copy and Reverse instead changed from nearly uniform correctness at the training length to nearly uniform incorrectness at the first unseen length.

4.2.2 Task-Specific Diagnostics

Position bins do not directly test whether errors are associated with the computation required by each task. Addition was therefore analysed according to incoming-carry condition, while Reverse was analysed according to the distance between an output position and its corresponding source position.

At the first unseen Addition length, the learned model produced 88.14% error on ordinary digits without an incoming carry and 89.09% error on digits with an incoming carry, a difference of 0.96 percentage points. For the sinusoidal model, the corresponding rates were 88.46% and 88.46%, differing by less than 0.01 percentage points. The optional leading result digit was excluded from this comparison.

The Addition collapse was therefore not concentrated on digits receiving an incoming carry. This does not demonstrate that a length-general carry procedure was learned. Instead, it shows that once extrapolation failed, ordinary digits were predicted poorly under both carry conditions.

For Reverse, the source position associated with output position i is $L - 1 - i$. The normalised dependency distance was calculated as

$$d_i = \frac{|(L-1-i) - i|}{L-1}.$$

At the training length, error was zero across all dependency-distance bins. At length 256, error varied only from approximately 98.54% to 98.92%. Shorter and longer reversal dependencies therefore failed at comparable rates. The failure was not restricted to positions requiring the longest input-output dependencies.

4.2.3 Attention as Descriptive Evidence

Attention statistics were computed at the training length, first unseen length, and longest evaluated length using the first five diagnostic batches. This corresponded to at most 40 Addition examples and five Delayed Copy or Reverse examples. The statistics are consequently treated as descriptive case studies rather than multi-seed estimates or causal explanations.

Normalised attention entropy increased for Addition and Reverse. For Addition with learned positional embeddings, it increased from 0.157 at length 16 to 0.278 at length 32 and 0.594 at length 1024. For Reverse, it increased from 0.539 at length 128 to 0.731 at length 256 and 0.910 at length 1024.

Delayed Copy did not follow the same pattern. Sinusoidal entropy changed from 0.645 at the training length to 0.627 at the first unseen length, while RoPE changed from 0.605 to 0.577. Both models nevertheless fell from approximately 100% to 0% exact-match accuracy. Their entropy values at length 1024 were 0.630 and 0.638, respectively.

Local attention ratio decreased with sequence length in every task. However, this metric used a fixed neighbourhood of $|i - j| \leq 10$, so part of the reduction follows mechanically from increasing the total sequence length. Average attention distance has a similar scale dependence because it is measured in token positions.

The attention results therefore do not support one universal attention-diffusion explanation. Increased entropy accompanied failure for Addition and Reverse, but Delayed Copy collapsed without a corresponding increase at the first unseen length. Attention weights are consequently used as supporting evidence rather than as a separate failure mechanism, consistent with the broader caution that attention patterns alone do not establish feature importance or causality [10].

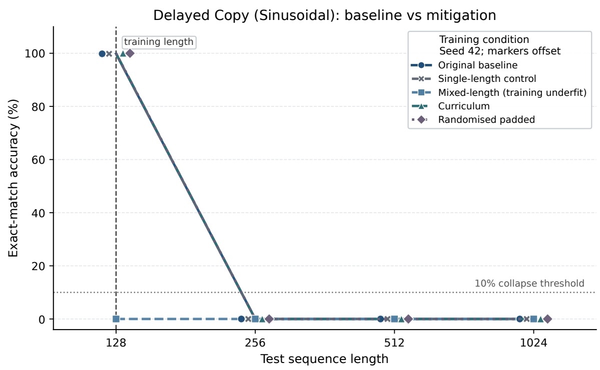

Figure 4.6: Original Delayed Copy sinusoidal baseline compared with the single-length control, mixed-length, curriculum, and randomised-padded conditions. Results use seed 42. Error bars are not shown because each line represents one matched experimental run.

4.3 Mitigation Results

The mitigation experiments tested specific hypotheses rather than treating all interventions as interchangeable. Mixed-length training tested whether simultaneous exposure to several lengths reduced specialisation to one fixed length. Curriculum training tested whether gradually increasing length addressed an optimisation barrier. Randomised padded training tested whether varying effective sequence length reduced dependence on fixed separator and output positions.

A mitigation was considered an interpretable extrapolation test only if it first preserved training-length exact-match accuracy. Variants below the 90% training-fit threshold were classified as underfitting rather than as failed extrapolation mitigations.

Figure 4.6 provides the most complete baseline-mitigation comparison. The single-length control reproduced the original baseline exactly: both achieved 99.8% at length 128 and 0% at every unseen length. This confirmed that the mitigation pipeline could reproduce the reference result when the training distribution was unchanged.

The mixed-length variant achieved 0% at the training length and was therefore classified as underfitting. Curriculum and randomised-padded training both achieved 100% at length 128, providing valid intervention tests, but neither improved exact-match accuracy at length 256 or beyond.

Condition	Train EM (%)	First OOD EM (%)	Interpretation
Addition learned: mixed-length v1	0.0	0.0	Training-length underfitting; extrapolation inconclusive.
Addition learned: mixed-length v2	0.0	0.0	Strengthened variant also underfit.
Copy sinusoidal: single-length control	99.8	0.0	Reproduced the original baseline.
Copy sinusoidal: mixed-length	0.0	0.0	Training-length underfitting; extrapolation inconclusive.
Copy sinusoidal: curriculum	100.0	0.0	Preserved training fit; no observed OOD improvement.
Copy sinusoidal: randomised padded	100.0	0.0	Preserved training fit; no observed OOD improvement.
Copy RoPE: curriculum	100.0	0.0	Preserved training fit; no observed OOD improvement.
Copy RoPE: randomised padded	100.0	0.0	Preserved training fit; no observed OOD improvement.

Table 4.3: Training-length and first-unseen-length exact-match (EM) accuracy for the evaluated mitigation conditions. All comparisons use seed 42.

The Reverse curriculum run also achieved 100% at length 128 and 0% at the unseen lengths. However, its vocabulary size differed from the original baseline, so it was not a fully matched comparison and is treated only as exploratory supporting evidence.

The mitigation conclusions are intentionally bounded. Mixed-length training caused underfitting in the evaluated configurations, while curriculum and randomised-padded training preserved training-length performance but produced no observed improvement at unseen lengths. These results do not establish that the failures are universally resistant to mitigation or that alternative architectures, optimisation settings, or length distributions could not improve performance.

4.4 Failure Taxonomy

The combined results support a taxonomy that distinguishes qualitatively different outcomes instead of describing every unsuccessful configuration as a length-generalisation failure.

The taxonomy is applied sequentially. Training fit is checked first, followed by seed stability. Only configurations that reliably learn the training task are evaluated for extrapolation collapse. Position-wise and task-specific diagnostics then describe the form of that collapse. Finally, mitigation outcomes are interpreted only when the intervention preserves the original training-length performance.

Attention change is not included as a separate category because the patterns were not consistent across tasks. It remains descriptive supporting evidence rather than a universal explanation of failure.

4.5 Exploratory Real-World Extension

The SelfRegulationSCP2 experiment provides an exploratory contrast between the controlled synthetic tasks and a real-world sequence-classification setting. The experimental design is described in Section 3.10.

Figure 4.7 shows mean classification accuracy of approximately 49–56% across the evaluated prefix lengths. Unlike the eligible synthetic baselines, no sharp decline occurs when the input length increases. However, performance at the training prefix is already close to the 50% uniform-prediction reference.

Category	Operational interpretation
Training-length underfitting	Training-length exact-match accuracy is below 90%. The original task was not learned sufficiently, so long-length performance cannot be interpreted as extrapolation failure.
Seed instability	Training success differs across the three random seeds. A pooled mean can therefore conceal qualitatively different successful and unsuccessful runs.
Immediate extrapolation collapse	A configuration satisfies multi-seed baseline eligibility but falls below 10% exact-match accuracy at the first evaluated unseen length. "Immediate" refers to the sampled evaluation grid rather than the unknown exact transition between two tested lengths.
Spatial error structure	Position-wise or task-conditioned token errors reveal how failure is distributed. Addition retains a modest positional gradient, whereas Delayed Copy and Reverse show almost uniform near-reference error.
No observed improvement under tested mitigation	A mitigation preserves training-length performance but does not improve first-unseen-length exact-match accuracy relative to its matched baseline. This conclusion applies only to the tested intervention and configuration.

Table 4.4: Failure taxonomy derived from baseline eligibility, multi-seed robustness, position-wise diagnostics, and mitigation experiments.

The resulting curve is therefore flat but weak and cannot be interpreted as evidence of successful length generalisation.

The differences between positional encodings are small relative to the overall performance level and variation across seeds, and no formal significance test was conducted. Moreover, classification accuracy is not directly comparable with synthetic exact-match accuracy: the classifier predicts one sequence-level label, whereas the synthetic tasks require every output token to be correct. Conclusions are consequently limited to this dataset and experimental setting.

This extension used one small dataset and a Transformer configuration chosen for comparability rather than task-specific optimisation. It did not include a specialised time-series model or a strong task-specific baseline. Accordingly, it demonstrates the importance of establishing an adequate real-world baseline before drawing conclusions about behaviour under changes in input length, rather than validating the synthetic failure taxonomy.

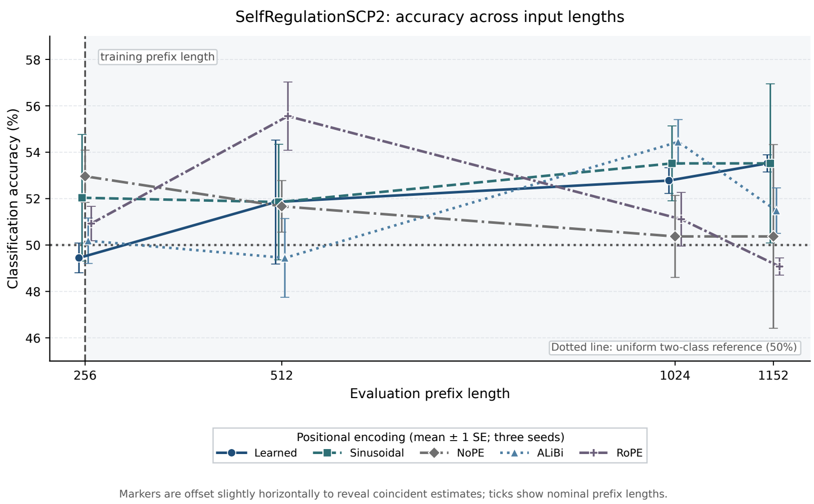

Figure 4.7: Mean SelfRegulationSCP2 classification accuracy across three seeds. Error bars show ±1 standard error. The dotted line denotes the 50% uniform two-class prediction reference, and the dashed line marks the training prefix length of 256.

4.6 Chapter Summary

Five task-encoding configurations reliably learned their training tasks, but all five collapsed at the first evaluated unseen length. Addition retained a modest position-wise error gradient, whereas Delayed Copy and Reverse produced almost uniform near-reference token error. Carry condition and dependency distance did not explain the respective Addition and Reverse failures, and attention behaviour did not provide a universal mechanism.

Mixed-length variants underfit in the evaluated settings. Curriculum and randomised-padded Delayed Copy variants preserved training-length performance but produced no observed improvement at unseen lengths. The exploratory SelfRegulationSCP2 experiment produced a flat but weak classification curve rather than the synthetic exact-match collapse. Together, these findings support the failure taxonomy developed in Table 4.4.

Chapter 5

Discussion

This chapter interprets the findings in relation to the primary research question and its five supporting questions. The central result is not simply that performance decreases at longer lengths. Instead, the experiments reveal several empirically different outcomes: failure to learn the training task, instability across seeds, collapse after a reliable training fit, different spatial distributions of token errors, and no observed improvement under particular tested mitigations.

These distinctions are important because configurations with the same long-length accuracy may provide very different evidence. A model that never learns the training task cannot be interpreted in the same way as one that moves from approximately 100% training-length accuracy to 0% at the first unseen length.

5.1 Answers to the Research Questions

5.1.1 RQ1: Which Configurations Reliably Learn the Training Task?

Five task-encoding configurations satisfied the baseline-eligibility criterion across all three seeds: Addition with learned and sinusoidal positional encodings, Delayed Copy with sinusoidal encoding and RoPE, and Reverse with learned positional embeddings. These configurations provide valid baselines for evaluating length extrapolation because they first demonstrated reliable training-length performance.

Other configurations produced different outcomes. Delayed Copy with learned positional embeddings and Reverse with sinusoidal encoding succeeded only for some seeds. NoPE, ALiBi, and several remaining task-encoding combinations did not reliably learn their training tasks under the evaluated architecture and optimisation settings. Their poor long-length performance is therefore classified as training-length underfitting rather than extrapolation collapse.

This result demonstrates why multi-seed evaluation is not merely a method for adding uncertainty bars. In some configurations, seed variation separates successful task learning from complete failure. Reporting only a pooled mean or a single selected seed could therefore produce a misleading account of model capability. This interpretation is consistent with broader concerns about the reproducibility of neural-network results across random initialisations [20].

5.1.2 RQ2: When Does Eligible Performance Collapse?

All five eligible configurations fell below the 10% exact-match threshold at the first evaluated unseen length. Addition collapsed when digit length increased from 16 to 32. Delayed Copy and Reverse collapsed when sequence length increased from 128 to 256. Exact-match accuracy then remained at 0% at every longer evaluated length.

This behaviour is classified as immediate extrapolation collapse. However, "immediate" refers to the sampled evaluation grid. The experiments show that collapse had occurred by lengths 32 and 256, but do not locate the exact transition between the training length and the first tested unseen length.

The result is consistent with evidence that Transformers can fit shorter instances while failing under length or formal-language shifts [27, 5, 1]. It also supports the view that extrapolation depends on whether optimisation discovers a computation whose structure remains valid as length changes [31]. Nevertheless, high training-length accuracy does not prove that a particular internal algorithm was learned.

The evidence establishes the narrower conclusion that the behaviour producing high training-length accuracy did not preserve complete sequence-level correctness at the first sampled longer length.

5.1.3 RQ3: Where Within the Output Do Errors Occur?

The position-wise analysis showed that 0% exact-match accuracy does not correspond to one common token-error pattern. At the first unseen length, Addition retained a modest positional gradient. Error was lower near the beginning of the output and approached the 90% uniform digit-prediction reference towards later positions. The sinusoidal model showed a larger position-wise range than the learned model.

Delayed Copy and Reverse instead produced token-error rates close to the 99% uniform valid-token reference throughout the output. These models changed from nearly uniform correctness at the training length to nearly uniform incorrectness at the first unseen length. Their failures were therefore more spatially distributed than the Addition failure.

The task-specific diagnostics further constrain the interpretation. Addition errors differed by less than one percentage point between ordinary digits with and without an incoming carry. The failure was therefore not primarily concentrated on carry-dependent digits. Similarly, Reverse error remained almost constant across dependency-distance bins, showing that the collapse was not restricted to output positions requiring the longest input–output mapping.

These findings show why exact-match accuracy should be accompanied by error-location diagnostics: concentrated errors may support task-specific analysis, whereas distributed near-reference error indicates that the whole longer output is unreliable. Addition retained limited positional structure, whereas Delayed Copy and Reverse exhibited distributed near-reference collapse.

These distinctions also have a practical implication for evaluating models that produce structured outputs. If errors are concentrated in particular regions, targeted checks may help identify vulnerable output positions. By contrast, distributed near-reference failure means that inspecting only a small number of positions could give misleading reassurance about the reliability of the complete output. Models expected to process sequences longer than those used during training should therefore be evaluated across both length and output position, rather than through aggregate accuracy alone. This is an implication for evaluation practice, not evidence that the specific synthetic error patterns necessarily occur in every deployed system.

5.1.4 RQ4: How Does Positional Encoding Affect the Outcome?

Positional encoding strongly affected whether the training task was learned, but no encoding was uniformly successful across all three tasks. Learned positional embeddings were eligible for Addition and Reverse but unstable for Delayed Copy. Sinusoidal encoding was eligible for Addition and Delayed Copy but unstable for Reverse. RoPE was eligible for Delayed Copy but underfit or only partially fit the other tasks.

The comparison identifies a boundary on what can be inferred from the positional mechanism alone. An encoding that can mathematically represent unseen positions did not necessarily enable reliable training fit or length extrapolation in the compact bidirectional encoder evaluated here. The experiments therefore do not support a universal ranking of positional encodings; they show that the observed outcome depends on the task, architecture, attention regime, representation, and optimisation procedure. This bounded interpretation is consistent with evidence that positional encoding does not determine length-generalisation behaviour independently of the surrounding experimental setting [11].

The comparison in Table 2.1 clarifies why these results do not directly contradict earlier findings. Press et al. [19] demonstrated train-short/test-long behaviour for ALiBi in causal language modelling, whereas ALiBi did not satisfy the training-length eligibility criterion on any of the three tasks evaluated here. This identifies a boundary on transfer across attention regimes, objectives, model scales, and task representations; it does not refute the original ALiBi result. Similarly, the strong NoPE behaviour reported for decoder-only Transformers by Kazemnejad et al. [11] does not imply that NoPE must fit the compact bidirectional sequence-to-sequence models studied here. The apparent disagreement therefore supports a conditional conclusion: positional methods cannot be ranked independently of the architecture and experimental setting in which they are evaluated.

Similarly, RoPE systematically incorporates position into query and key representations [24], but that representation does not guarantee that optimisation will discover a length-general task solution. Conversely, recent arithmetic studies show that task-aligned digit representations can substantially improve long-length addition [13, 15]. This reinforces the narrower interpretation that positional behaviour depends on task representation and architecture, rather than establishing a universal ranking of positional encoding methods.

5.1.5 RQ5: Do the Tested Mitigations Improve Extrapolation?

The mitigation experiments produced two different outcomes. Mixed-length training underfit in the evaluated Addition and sinusoidal Delayed Copy settings. Because these variants did not learn the training task, their long-length performance is inconclusive as evidence about extrapolation.

Curriculum and randomised-padded training preserved training-length performance for the eligible Delayed Copy configurations. However, both conditions achieved 0% exact-match accuracy at the first unseen length, matching the original baselines. These are valid negative results: the interventions preserved the original task but produced no observed extrapolation improvement in the tested configurations.

This conclusion must remain specific to the interventions evaluated. Curriculum learning covers a broad family of approaches for ordering examples by difficulty [4]; this project tested only one four-stage length curriculum using a representative seed. The result cannot establish that curriculum learning is generally ineffective.

The randomised-padded condition must also be distinguished from randomised positional encoding. It used shuffled, variable-length examples padded to a common maximum length, but did not sample positional indices from a wider range. It therefore tested a different hypothesis from the randomised positional strategy proposed by Ruoss et al. [21]. The appropriate conclusion is consequently "no observed improvement under tested mitigation", not the stronger claim that the failure is resistant to all mitigation.

5.2 Interpretive Contribution of the Taxonomy

The main interpretive contribution is a five-category taxonomy:

1. training-length underfitting;
2. seed instability;
3. immediate extrapolation collapse;
4. spatial error structure; and
5. no observed improvement under tested mitigation.

These categories are not proposed as mutually exclusive causal mechanisms. Instead, they form a sequential framework for evaluating experimental evidence. Training-length performance is checked first because extrapolation cannot be assessed before the original task is learned. Seed consistency is then examined because an average may conceal successful and unsuccessful runs. Eligible configurations are evaluated across unseen lengths, after which token-level diagnostics characterise the distribution of failure. Finally, mitigation is judged only when the intervention preserves the training baseline.

The contribution is not a new causal theory of Transformer failure or an exhaustive taxonomy of every possible failure mechanism. Instead, it is an evidence-organising procedure that prevents superficially similar long-length results from receiving the same interpretation. For example, a NoPE or ALiBi model that never learns the training task and an eligible model that moves from approximately 100% to 0% exact-match accuracy may have identical performance at the longest length, but they provide fundamentally different evidence. The taxonomy makes that distinction explicit and reproducible.

Attention behaviour was not retained as a separate taxonomy category. Normalised entropy increased for Addition and Reverse but remained broadly stable for Delayed Copy at the first unseen length. Attention diffusion therefore did not provide a universal explanation. More generally, attention weights should not automatically be treated as causal explanations of model predictions [10]. In this dissertation, they provide descriptive supporting evidence only.

5.3 Limitations and Interpretation Boundaries

The synthetic benchmark uses one compact Transformer architecture, three tasks, five positional encoding implementations, and three random seeds. These choices enable controlled comparison but limit the extent to which the results can be generalised to larger models, decoder-only architectures, different attention mechanisms, or other algorithmic tasks.

The position-wise, task-specific, attention, and mitigation diagnostics use representative seed-42 checkpoints. They describe the form of failure in selected models rather than establishing that every seed fails in precisely the same way. Attention statistics are also averaged over a limited number of diagnostic batches and are not causal tests.

The first failure length is measured on a discrete evaluation grid. A denser set of test lengths would be required to locate the transition more precisely. Exact match is also deliberately strict: one incorrect token makes the entire prediction incorrect. Token-error percentages and position-wise profiles were therefore necessary to reveal partial and spatially structured behaviour.

The mitigation experiments were targeted tests rather than exhaustive hyperparameter searches. Some baseline and mitigation configurations also used different batch sizes. The Delayed Copy single-length control reproduced its original baseline, supporting that particular pipeline, but equivalent controls were not completed for every task–encoding pair. The Reverse curriculum run additionally used a different vocabulary size and was therefore treated as exploratory rather than as a fully matched comparison.

Finally, the SelfRegulationSCP2 extension uses one small dataset and does not establish strong held-out performance at the training prefix length. Its flat accuracy curve cannot be interpreted as successful real-world length generalisation. Classification accuracy and synthetic exact-match accuracy also measure fundamentally different output requirements.

These limitations bound the conclusions but do not invalidate the controlled findings. Within the evaluated setting, the evidence consistently shows that reliable training-length success did not transfer to the first sampled longer length and that the observable form of failure differed across tasks.

5.4 Implications

The results suggest that length-generalisation studies should proceed in a defined order:

1. establish that the training task is reliably learned;
2. evaluate multiple seeds before pooling results;
3. report a curve over increasing unseen lengths;
4. supplement exact match with token-level and position-wise errors;
5. apply diagnostics specific to the task computation; and
6. compare interventions with matched reference baselines.

This procedure distinguishes long-context capacity from length-general computation. A model may technically process a longer input without preserving the rule that produced correct predictions at the training length. Likewise, a positional encoding designed for longer contexts does not by itself guarantee algorithmic extrapolation.

Future work should evaluate more seeds and denser length grids, use matched optimisation settings, and test additional architectures and tasks. Direct randomisation of positional indices [21], intermediate-step or scratchpad representations [1, 11], and architectures that support iterative computation provide particularly relevant directions. A stronger real-world extension should first establish a competitive training-prefix baseline before examining performance under controlled length shifts.

5.5 Summary

The primary research question can therefore be answered as follows: length-generalisation failure in the evaluated Transformers appears as a set of distinguishable empirical behaviours rather than one uniform decrease in accuracy. Some configurations underfit, some are unstable across seeds, and five reliably learn their training tasks before collapsing at the first unseen length. Addition retains a modest positional error gradient, while Delayed Copy and Reverse show almost uniform near-reference collapse. The tested valid mitigations preserve training performance but produce no observed improvement beyond it.

Beyond this benchmark, the contribution is methodological: the taxonomy gives practitioners a structured sequence of checks for distinguishing underfitting, seed instability, and extrapolation collapse before relying on a model at unseen lengths.

Chapter 6

Conclusion

This dissertation investigated how length-generalisation failure appears when compact Transformer models trained on short sequences are evaluated on substantially longer sequences. The study used three controlled synthetic tasks, five positional encoding settings, three random seeds, position-wise and task-specific diagnostics, targeted mitigation experiments, and an exploratory real-world extension.

The principal conclusion is that high training-length accuracy is not sufficient evidence that a Transformer has learned a length-general computation. Every task–encoding configuration that satisfied the multi-seed baseline-eligibility criterion collapsed at the first evaluated unseen length. However, the wider experiment also demonstrated that poor long-length performance does not always represent the same phenomenon. Training-length underfitting, seed instability, extrapolation collapse, and different spatial error structures require separate interpretations.

6.1 Principal Conclusions

Five configurations reliably learned their training task across all three seeds: Addition with learned and sinusoidal positional encodings, Delayed Copy with sinusoidal encoding and RoPE, and Reverse with learned positional embeddings. These configurations provided valid baselines for examining length extrapolation.

All five fell to 0% exact-match accuracy at the first evaluated unseen length. Addition collapsed when digit length increased from 16 to 32, while Delayed Copy and Reverse collapsed when sequence length increased from 128 to 256. Performance remained at 0% exact match at every longer tested length. Within the sampled evaluation grid, the eligible models therefore exhibited immediate extrapolation collapse rather than gradual degradation.

No positional encoding was uniformly successful across the three tasks. Learned, sinusoidal, and rotary encodings produced task-dependent combinations of reliable training fit, seed instability, and extrapolation collapse. NoPE and ALiBi frequently underfit under the evaluated architecture and optimisation settings. These outcomes should not be interpreted as universal rankings of the positional encoding methods; they apply to the specific bidirectional encoder architecture, task representations, and training procedure used in this dissertation.

The token-level diagnostics showed that the form of collapse differed across tasks. At the first unseen length, Addition produced approximately 86.9–88.0% token error and retained a modest positional gradient, with lower error near the beginning of the output. Delayed Copy and Reverse produced approximately 98.8–99.0% token error that was almost uniform throughout the output.

Task-specific analysis further narrowed these interpretations. Addition errors differed by less than one percentage point between ordinary digits with and without an incoming carry. Reverse errors were similarly high across short and long dependency-distance bins. Addition failure was therefore not primarily concentrated on carry-dependent digits, while Reverse failure was not restricted to the longest positional mappings.

Attention statistics did not provide one universal explanation. Normalised attention entropy increased for Addition and Reverse but remained broadly stable for Delayed Copy at the first unseen length. Attention behaviour was therefore retained as descriptive supporting evidence rather than treated as a causal explanation or an independent failure category.

The mitigation experiments produced two outcomes. Mixed-length variants underfit in the evaluated Addition and sinusoidal Delayed Copy settings, making their extrapolation results inconclusive. Curriculum and randomised-padded Delayed Copy variants preserved training-length performance but produced no observed exact-match improvement at unseen lengths. These are bounded results for the interventions and configurations tested; they do not show that the failures are universally resistant to mitigation.

The SelfRegulationSCP2 extension produced mean classification accuracy of approximately 49–56% across the evaluated prefix lengths. It did not exhibit the catastrophic exact-match collapse observed on the synthetic tasks. However, accuracy was already close to the 50% uniform two-class reference at the training prefix length. The result therefore represents a flat but weak classification curve, not successful real-world length generalisation.

6.2 Achievement of the Project Aim

The project aim was to determine how Transformer length-generalisation failure appears across controlled sequence tasks and whether positional encoding or training interventions alter that behaviour. Table 6.1 summarises the stage reached for each component of this aim.

Project objective	Stage reached	Outcome
Establish reliable training baselines	Achieved	Multi-seed eligibility separated reliable training fit from underfitting and seed-dependent success.
Compare positional encodings	Achieved within scope	Five encoding settings were compared across three tasks and three seeds; no encoding was uniformly successful.
Identify failure behaviour	Achieved	Exact-match, token-error, position-wise, carry-conditioned, and dependency-distance analyses distinguished different forms of collapse.
Evaluate mitigation strategies	Achieved with bounded conclusions	Mixed-length, curriculum, and randomised-padded conditions were compared with original baselines; valid interventions produced no observed OOD improvement.
Explore a real-world setting	Exploratory stage reached	The SelfRegulationSCP2 extension was completed, but its weak starting baseline prevents strong conclusions about real-world length generalisation.

Table 6.1: Stage reached relative to the principal project objectives.

The project therefore achieved its central experimental and diagnostic aims. It did not identify a successful mitigation or establish a strong real-world length-generalisation result. These are empirical outcomes rather than missing parts of the analysis: the interventions were evaluated, but their results were either invalidated by underfitting or showed no observed improvement after preserving the training baseline.

6.3 Contributions

The dissertation makes four principal contributions.

1. It establishes a baseline-eligibility procedure that prevents training-length underfitting from being misclassified as extrapolation failure.
2. It provides a controlled multi-task, multi-encoding, and multi-seed evaluation of Transformer length behaviour under a common train-short/test-long protocol.
3. It moves beyond aggregate accuracy by introducing percentage-based token errors, position-wise profiles, carry-conditioned errors, and dependency-distance analysis to describe where and how failure occurs.
4. It proposes an evidence-based taxonomy comprising training-length underfitting, seed instability, immediate extrapolation collapse, spatial error structure, and no observed improvement under tested mitigation.

The taxonomy is not intended as an exhaustive theory of every possible Transformer failure. Its contribution is methodological: it provides an ordered set of checks that makes experimental claims more precise and prevents distinct outcomes from being reduced to the statement that a model simply "fails at longer lengths".

6.4 Limitations and Open Questions

The conclusions are limited to one compact Transformer architecture, three synthetic tasks, five positional encoding implementations, and three baseline seeds. Larger models, causal or decoder-only architectures, alternative task representations, and different optimisation procedures may produce different behaviour.

The detailed position-wise, attention, task-specific, and mitigation analyses use representative seed-42 checkpoints. They characterise selected failures but do not establish the multi-seed prevalence of every diagnostic pattern. The first failure point is also bounded by the discrete evaluation grid; the precise transition between the training length and first tested unseen length remains unknown.

The mitigation experiments were targeted rather than exhaustive hyperparameter searches. Some optimisation settings differed between original and mitigation runs, although the Delayed Copy single-length control reproduced its original baseline. The Reverse curriculum experiment also used a different vocabulary size and was consequently treated as exploratory rather than as a fully matched comparison.

These limitations leave several open questions. It remains unclear whether the same position-wise patterns occur across all successful seeds, whether collapse begins immediately above the training length, and whether alternative representations or iterative architectures would learn more transferable solutions. It is also unresolved whether carefully tuned mixed-length training can avoid the underfitting observed here.

6.5 Future Work

First, diagnostics and valid mitigations should be repeated across more seeds with matched batch sizes, vocabularies, optimisation settings, and single-length controls. A denser grid between the training length and first observed collapse would locate the transition more precisely.

The second priority is to test interventions that target different hypotheses. Directly randomised positional indices would provide a closer test of positional exposure than right-padded variable-length training. Architectures with recurrent or iterative computation provide another direction because they introduce computational depth beyond a fixed stack of Transformer blocks [7, 29]. Structured intermediate computation offers a complementary direction for algorithmic extrapolation [1, 9].

A stronger mechanistic analysis should move beyond attention summaries. Controlled head and layer ablations, attention replacement, probing, and causal interventions could test whether particular internal components contribute directly to the observed output errors.

Finally, a stronger real-world extension should first establish a competitive held-out baseline at the training prefix length, then evaluate specialised time-series models, additional datasets and seeds, and appropriate statistical comparisons. More broadly, the taxonomy could inform benchmark design and practical model validation by requiring distinct failure behaviours to be identified before performance at unseen lengths is trusted.

6.6 Final Conclusion

Transformer length-generalisation failure in this study appeared as a set of empirically distinguishable behaviours rather than one uniform decrease in accuracy. A meaningful evaluation must first establish that the original task was learned, determine whether that result is stable across seeds, locate the first observed extrapolation collapse, examine how errors are distributed, and test interventions against a valid reference baseline.

Within the evaluated synthetic benchmark, every eligible configuration collapsed at the first sampled unseen length. The surrounding evidence, however, showed that the form and interpretation of failure varied across tasks and experimental conditions. The central contribution of this dissertation is therefore not only the observation that length extrapolation failed, but a reproducible framework for determining what kind of failure occurred and what conclusions the available evidence can legitimately support.

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Appendix A

Supplementary Experimental Results

Chapter 4 presents the figures required to support the principal findings. This appendix provides additional error profiles, task-specific diagnostics, and mitigation comparisons that support the main analysis without being necessary for its continuous argument.

Unless otherwise stated, the detailed diagnostic figures use representative seed-42 checkpoints. They characterise selected eligible models and should not be interpreted as multi-seed estimates. Multi-seed baseline eligibility is established separately in Section 4.1.

A.1 Position-Wise Error Across All Lengths

The main chapter compares training-length and first-unseen-length error profiles. Figures A.1 A.3 extend that comparison across every evaluated length. Target positions are divided into ten relative bins, and all heatmaps use a common 0–100% token-error scale.

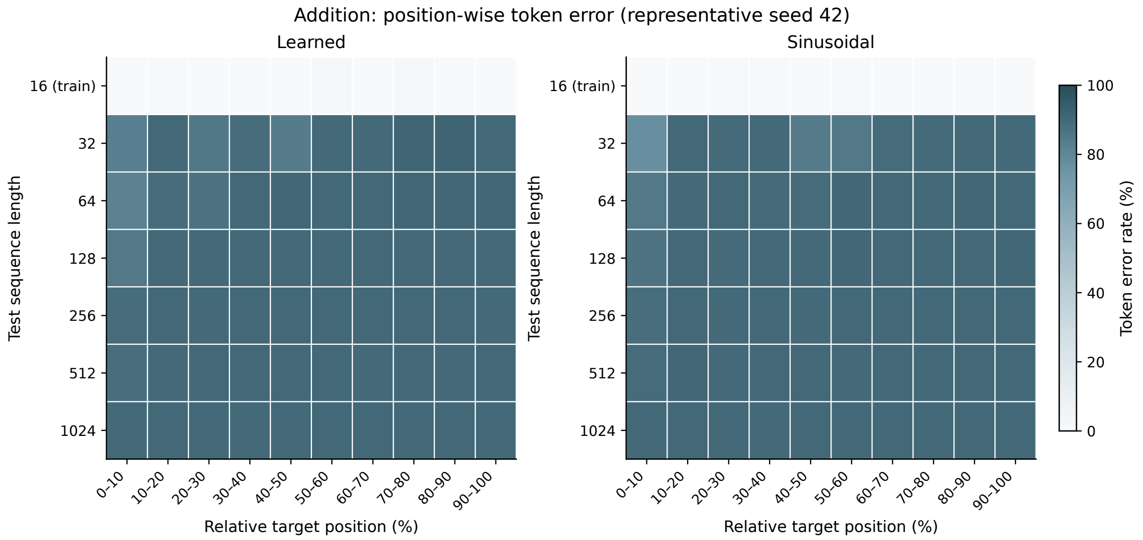

Figure A.1: Position-wise Addition token-error rates across all evaluated digit lengths for the learned and sinusoidal seed-42 checkpoints. Target positions are divided into ten relative-position bins.

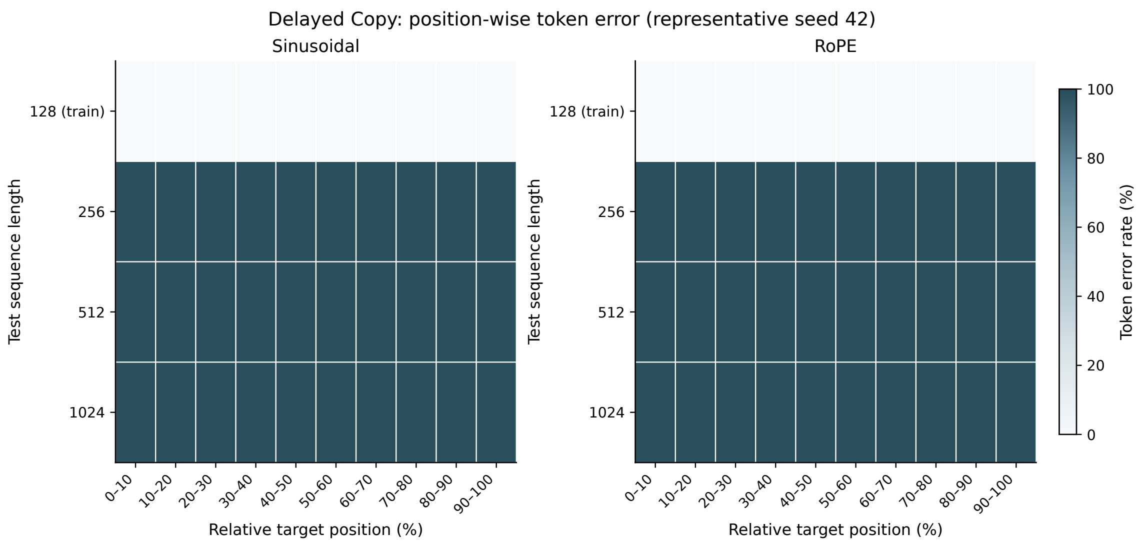

Figure A.2: Position-wise Delayed Copy token-error rates across all evaluated sequence lengths for the sinusoidal and RoPE seed-42 checkpoints. Target positions are divided into ten relative-position bins.

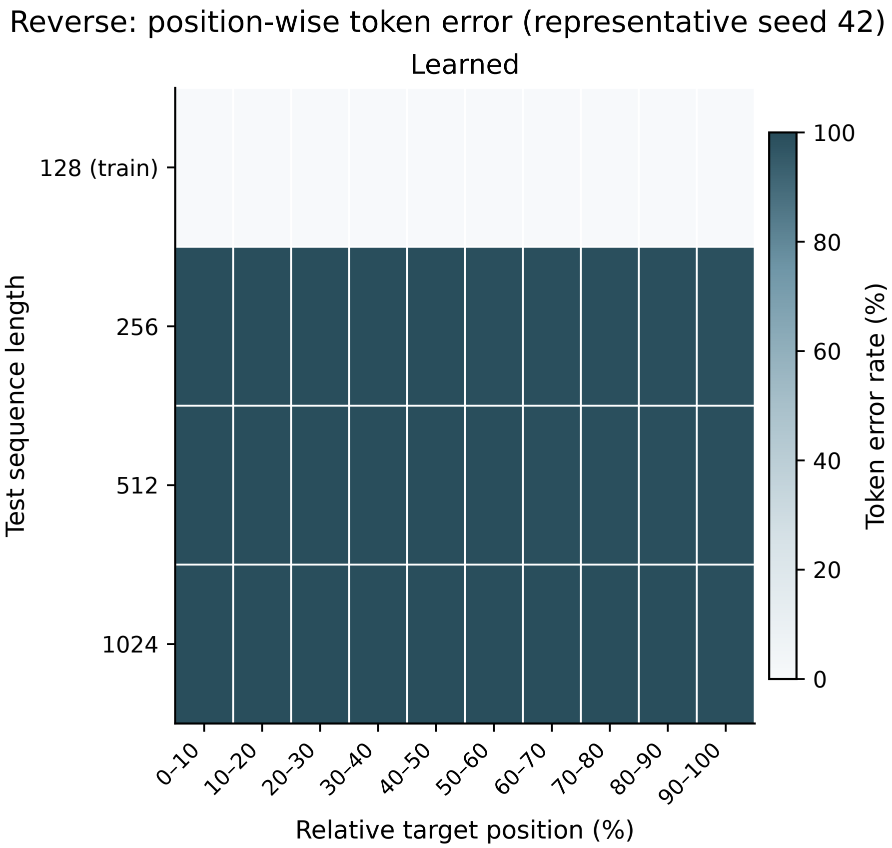

Figure A.3: Position-wise Reverse token-error rates across all evaluated sequence lengths for the learned seed-42 checkpoint. Target positions are divided into ten relative-position bins.

A.2 Additional First-Unseen Position Profiles

The Addition training-versus-first-unseen comparison is retained in the main chapter. Figures A.4 and A.5 provide the corresponding Delayed Copy and Reverse comparisons.

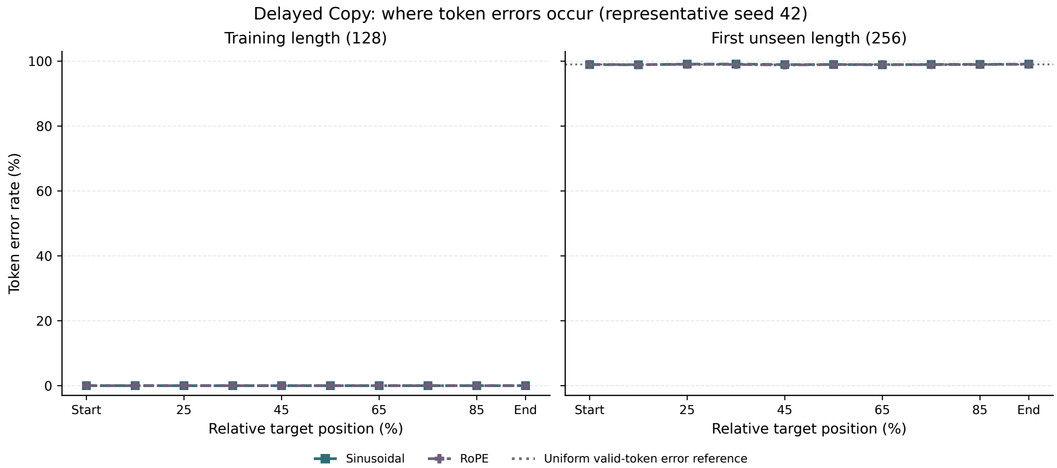

Figure A.4: Delayed Copy position-wise token-error rates at the training length and first evaluated unseen length for the sinusoidal and RoPE seed-42 checkpoints. The dotted line denotes the 99% uniform valid-token error reference.

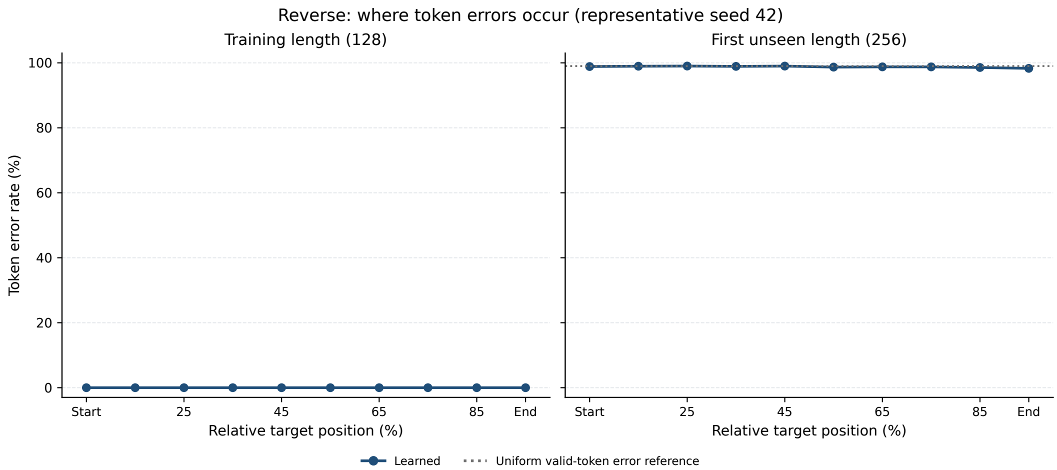

Figure A.5: Reverse position-wise token-error rates at the training length and first evaluated unseen length for the learned seed-42 checkpoint. The dotted line denotes the 99% uniform valid-token error reference.

A.3 Task-Specific Failure Diagnostics

The position-wise profiles were supplemented by diagnostics specific to the computation required by Addition and Reverse. Figure A.6 conditions Addition errors on the presence of an incoming carry. Figure A.7 groups Reverse errors by normalised source-target dependency distance.

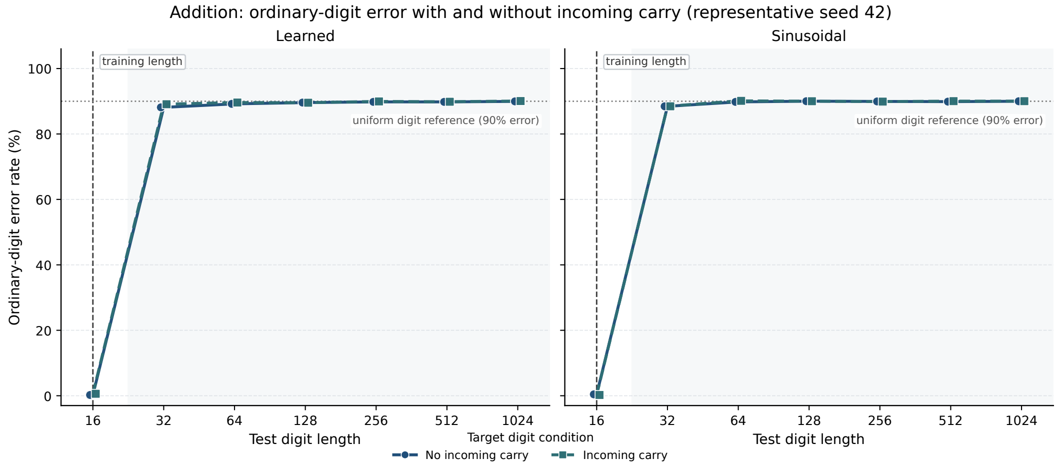

Figure A.6: Addition error rates for ordinary target digits with and without an incoming carry. The learned and sinusoidal seed-42 checkpoints are shown across all evaluated lengths. The optional leading result digit is excluded, and the dotted line denotes the 90% uniform digit-error reference.

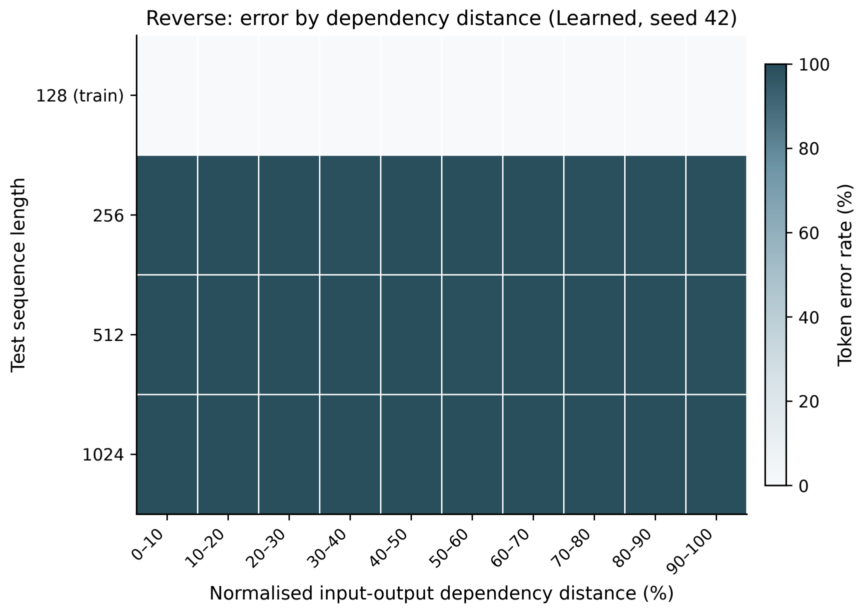

Figure A.7: Reverse token-error rates grouped by normalised source–target dependency distance for the learned seed-42 checkpoint. The approximately uniform error at unseen lengths shows that failure was not confined to the longest dependencies.

A.4 Error Decomposition Across Evaluation Lengths

The following figures report exact-match error and token error across the complete length schedule. They complement the exact-match degradation curves in the main chapter by showing the magnitude of token-level failure as a percentage.

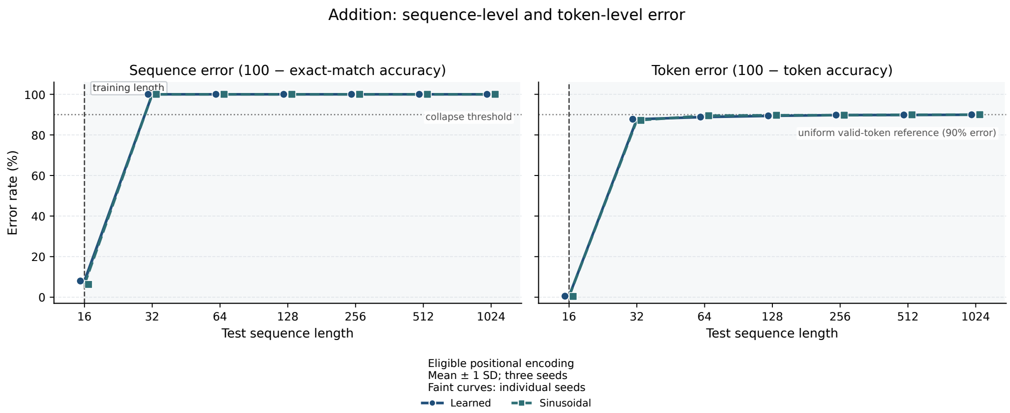

Figure A.8: Exact-match and token-error percentages across evaluation lengths for eligible Addition baselines.

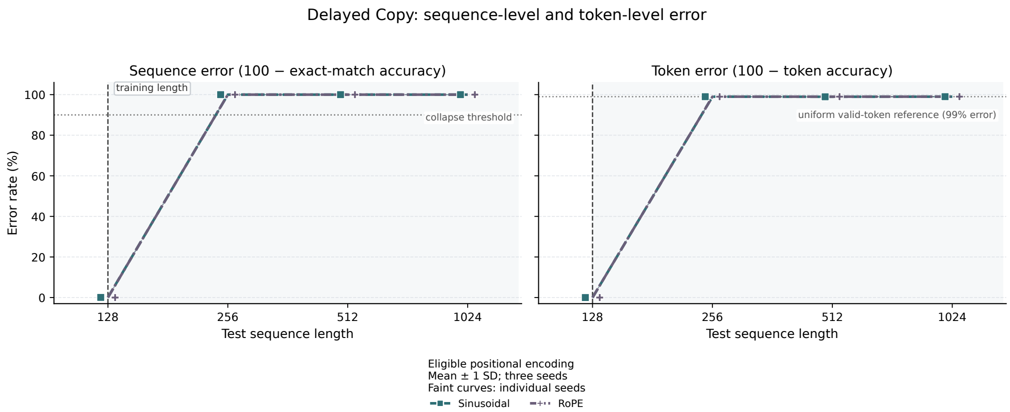

Figure A.9: Exact-match and token-error percentages across evaluation lengths for eligible Delayed Copy baselines.

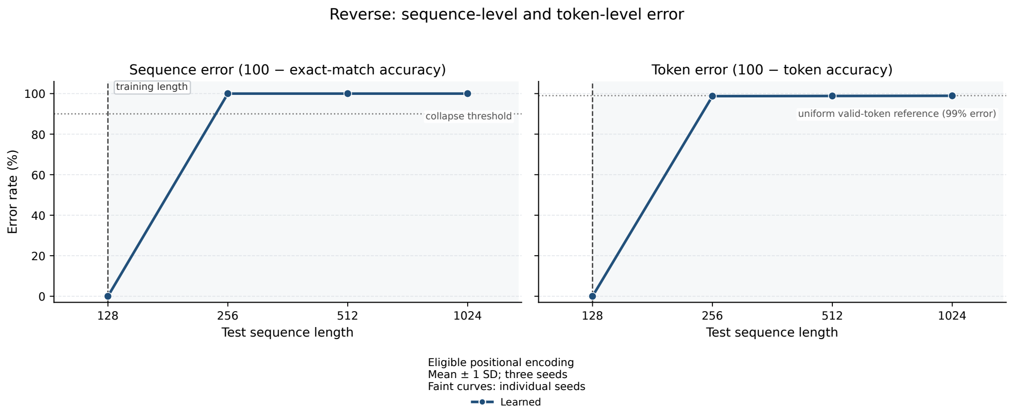

Figure A.10: Exact-match and token-error percentages across evaluation lengths for the eligible Reverse baseline.

A.5 Additional Mitigation Comparisons

The main chapter retains the Delayed Copy sinusoidal comparison because it contains the original baseline, single-length control, mixed-length, curriculum, and randomised-padded conditions. The remaining mitigation plots are provided here.

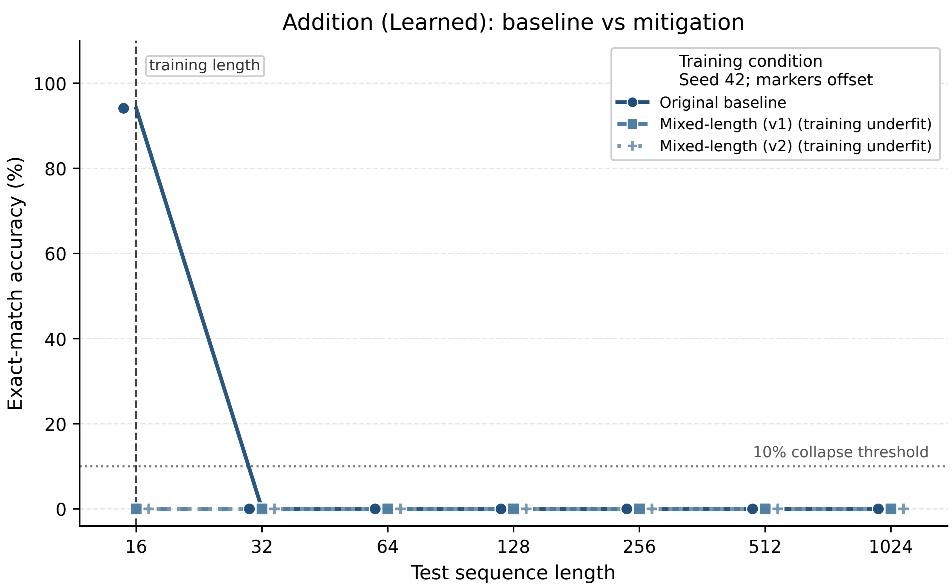

Figure A.11: Original Addition learned baseline compared with two mixed-length variants. Both interventions underfit at the training length, so their unseen-length results are inconclusive as extrapolation tests. Results use seed 42.

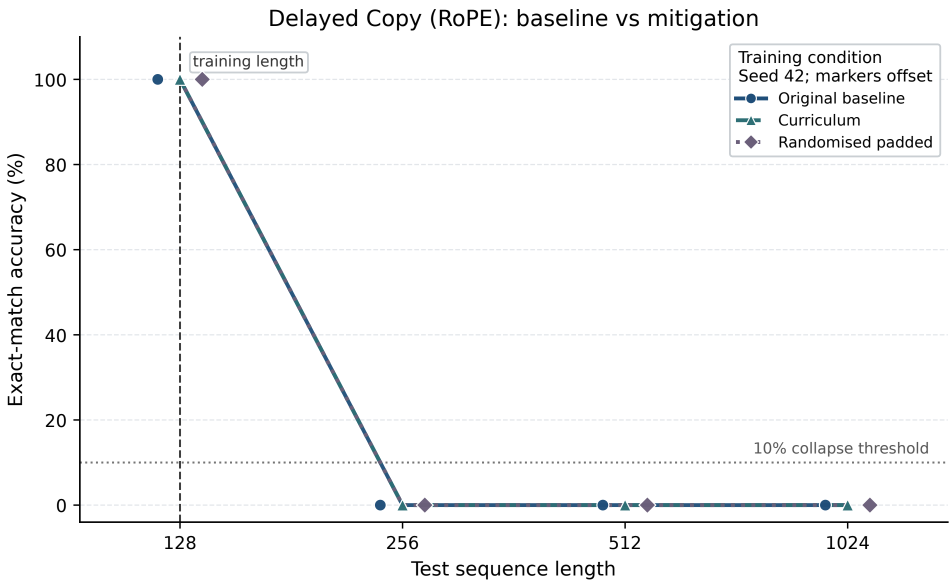

Figure A.12: Original Delayed Copy RoPE baseline compared with curriculum and randomised-padded training. Both interventions preserve training length performance but produce no observed improvement at unseen lengths. Results use seed 42.

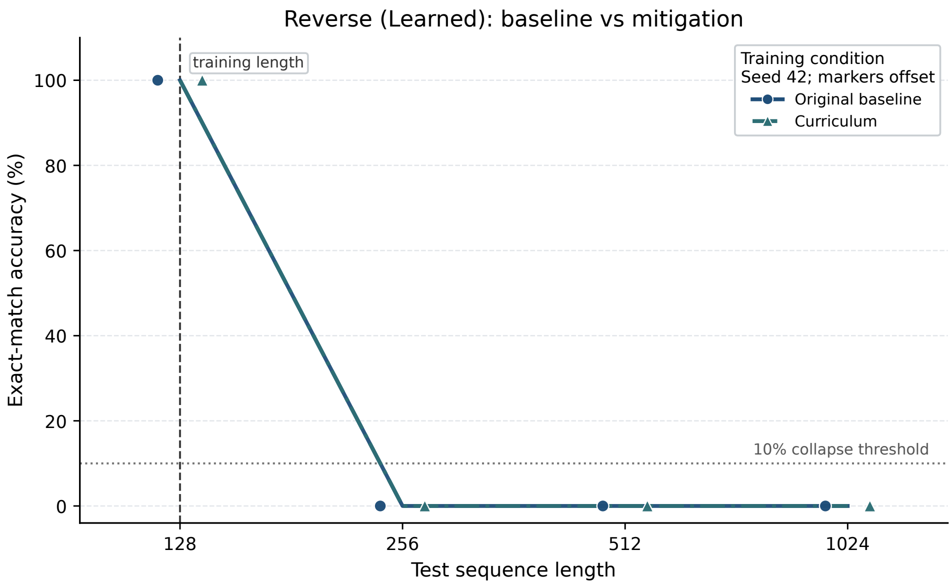

Figure A.13: Original Reverse learned baseline compared with the exploratory curriculum condition. The curriculum vocabulary differed from the original baseline, so this is not treated as a fully matched intervention comparison. Results use seed 42.

A.6 Interpretive Scope

The supplementary figures do not introduce additional statistical claims. They preserve the complete polished diagnostic record underlying the more focused comparisons in Chapter 4. The heatmaps and task-specific plots support the distinction between the modest positional gradient observed in Addition and the almost uniform near-reference failure observed in Delayed Copy and Reverse.

The mitigation figures likewise preserve the distinction between training-length underfitting and a valid intervention that retains training fit but produces no observed improvement at unseen lengths.

Appendix B

Supplementary Methodological Documentation

This appendix preserves detailed experimental documentation that supports reproducibility but is not required for the continuous methodological argument in Chapter 3. It records the complete experiment matrix, experimental controls, diagnostic scope, and repository traceability.

B.1 Baseline Experiment Matrix

Experiment component	**Values**	**Models**
Synthetic tasks	Addition, Delayed Copy, Reverse	–
Positional encoding conditions	Learned, sinusoidal, NoPE, ALiBi, RoPE	–
Synthetic seeds	42, 123, 2024	–
Synthetic baseline matrix	3 tasks $\times$ 5 encodings $\times$ 3 seeds	45
Real-world baseline matrix	1 dataset $\times$ 5 encodings $\times$ 3 seeds	15
Total baseline models	Synthetic and real-world experiments	60
Mitigation experiments	Targeted seed-42 follow-up conditions	Separate

Table B.1: Complete baseline experimental matrix. Model counts refer to independently trained checkpoints and exclude targeted mitigation runs.

B.2 Diagnostic Scope

Analysis	**Experimental scope**	**Interpretive purpose**
Baseline eligibility	All 45 synthetic baseline runs; three seeds per task–encoding condition	Separates stable training fit, seed instability, and underfitting.
Extrapolation curves	All three seeds for each eligible configuration	Locates collapse while preserving seed-level variation.
Position-wise errors	Five eligible seed-42 checkpoints; 1,000 test examples per length	Identifies whether errors concentrate at particular output positions.
Addition carry diagnostic	Eligible Addition seed-42 checkpoints; all evaluated target digits	Compares errors with and without an incoming carry.
Reverse distance diagnostic	Eligible Reverse seed-42 checkpoint; all evaluated target positions	Tests variation with normalised source–target distance.
Attention statistics	Five eligible seed-42 checkpoints; first five batches at selected lengths	Provides representative descriptive attention evidence.
Mitigation comparison	Targeted seed-42 interventions and corresponding seed-42 baselines	Tests whether interventions retain training fit and improve OOD performance.
Real-world extension	Five positional encodings and three seeds	Examines classification accuracy across input-prefix lengths.

Table B.2: Experimental scope and interpretive purpose of each analysis. Multi-seed conclusions are separated from representative seed-42 diagnostics.

B.3 Experimental Controls

Potential confound	Experimental control	Interpretive consequence
Different sampled data	Task–seed combinations used identically generated examples across positional encoding conditions.	Within-task differences are not explained by different sampled datasets.
Architecture or optimisation	Architecture, epochs, optimiser, learning rate, vocabulary, and length schedule were fixed within each task.	The positional encoding was the principal changed model condition.
Failure to learn the task	Eligibility required all three seeds to reach 90% training-length exact match.	Underfitting and seed instability are separated from extrapolation collapse.
Random-seed dependence	Every baseline condition used seeds 42, 123, and 2024.	Stable baseline conclusions require agreement across runs.
Padding dominance	Padding targets were excluded from loss, token accuracy, and exact match.	Reported performance concerns meaningful target tokens.
Test-set model selection	The final-epoch checkpoint was used; validation and test results did not select the checkpoint.	Longer-length results did not influence training or selection.
Diagnostic inconsistency	Position-wise evaluation recomputed aggregate accuracy before calculating error profiles.	Detailed diagnostics correspond to the ordinary evaluation results.
Mitigation pipeline changes	A single-length Delayed Copy control reproduced the mitigation pipeline without changing length exposure.	Pipeline effects can be distinguished from the tested intervention.
Mechanistic overinterpretation	Attention statistics were restricted to descriptive evidence.	Attention change is not presented as proof of a causal mechanism.

Table B.3: Experimental controls supporting interpretation of the baseline, diagnostic, and mitigation results.

B.4 Repository Traceability

Component	Repository location	Role
Synthetic task generation	`src/tasks/`	Implements Addition, Delayed Copy, Reverse, and dataset serialisation.
Dataset loading	`src/data/`	Loads synthetic JSONL records and prepares time-series prefixes.
Transformer models	`src/models/`	Implements attention, positional conditions, Transformer blocks, and the time-series classifier.
Training and evaluation	`src/training/`	Trains checkpoints and evaluates configured sequence lengths.
Baseline configurations	`configs/multiseed/`	Records task, architecture, training, encoding, seed, and output settings.
Mitigation configurations	`configs/mitigation/`	Records mixed-length, curriculum, randomised-padded, and control conditions.
Real-world configurations	`configs/realworld/`	Records the five positional conditions and three real-world seeds.
Raw results	`outputs/results/`	Preserves seed-level baseline, mitigation, and real-world measurements.
Derived diagnostics	`outputs/analysis/`; `position_diagnostics_aws/final`	Stores attention, position-wise, carry, dependency-distance, and mitigation summaries.
Report figures	`outputs/plots/final/`	Contains the consistently styled report figures.

Table B.4: Traceability between the experimental methodology and repository implementation.

Appendix C

Repository Guide

The implementation, experiment configurations, tests, result-processing scripts, and report-figure generation code used in this dissertation are available in the following public GitHub repository:

Project GitHub Repository

C.1 Repository Contents

The repository separates experiment definitions from generated outputs. Synthetic task implementations are stored in `src/tasks/`; dataset loading and tokenisation are in `src/data/`; and Transformer, positional-encoding, and time-series models are in `src/models/`. Training, evaluation, diagnostic, aggregation, and plotting programs are contained in `src/training/`.

Experiment settings are recorded as YAML files under `configs/`. The `multiseed/`, `mitigation/`, and `realworld/` subdirectories respectively contain the synthetic baseline, targeted intervention, and SelfRegulationSCP2 configurations. Raw evaluation CSV files are separated from derived summaries and final figures under `outputs/results/`, `outputs/analysis/`, and `outputs/plots/final/`.

C.2 Reproduction Entry Points

The repository's `README.md` provides environment-setup instructions, the expected directory structure, and commands for dataset generation, training, evaluation, baseline analysis, mitigation comparison, position-wise diagnostics, attention summaries, real-world evaluation, and final figure generation.

The principal execution sequence is:

```
configs/ -> task generation -> training -> evaluation
         -> raw result CSVs -> derived analysis -> final figures
```

Experiments are controlled through configuration files rather than hard-coded command-line settings. Each configuration records the task, sequence lengths, positional encoding, random seed, model hyperparameters, data location, checkpoint location, and result location.

C.3 Reproducibility Scope

The repository preserves the three-seed baseline configurations used for the main conclusions and the targeted seed-42 configurations used for detailed diagnostics and mitigation experiments. The dissertation distinguishes these two evidential scopes: multi-seed results determine baseline eligibility, whereas representative checkpoint analyses characterise how selected eligible models fail.

Large experiments may require a CUDA-enabled GPU. The final report figures can be regenerated from the retained result and summary files without retraining the complete experiment matrix. Exact commands and prerequisites are documented in the repository README.